Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Machine Learning Only Alert Types

This article contains only the machine learning alert types. They are listed alphabetically by their display name.

For the rule-based only alert types, see Rule-Based Only Alert Types.

For all the alert types, see All Alert Types Details.

Abnormal Parent / Child Process

A process that typically launches a small, consistent number of child processes has launched a new child process. Investigate the new child process or the parent process to see if it is benign.

This alert type has the following subtype categories:

- Machine Learning Anomaly Detection
- Rule Based Detection

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR Process Relationship Anomaly (XT1002)
- Tags: [Process Anomaly]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `parent_child`.

Severity

25

Alert Subtype: Machine Learning Anomaly Detection

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `machine_learning_anomaly_detection`.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- parent_proc_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `process_name` — name of the process
- `parent_proc_name` — name of the parent process
- `hostip` — host IP address
- `hostip_host` — host name
- `stability` — score measuring the time since the parent process launched the last child process
- `days_stable` — time since the parent process launched the last child process
- `diversity` — score measuring the number of child processes that the parent process spawned
- `child_count` — number of child processes that the parent process spawned

Use Case with Data Points

Each pair of parent/child processes (`parent_proc_name` and `process_name`) is examined periodically. If a parent process (`parent_proc_name`) with a small number of child processes (`diversity`, `child_count`) has not launched a new child process (`process_name`) for a long time (`stability`, `days_stable`) launches a new child process from a host (`srcip_host`), an alert is triggered.

Alert Subtype: Rule Based Detection

The Parent/Child Suspicious Process Creation rules are used to identify suspicious activity with parent/child relationships associated with process creation. Any one or more of these will trigger the Parent/Child Suspicious Process Creation alert types.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip
- stellar.rule_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name
- `stellar.rule_id` — Stellar Cyber rule ID

Link to Rule-Based Alert Types

Rules Contributing to Parent/Child Suspicious Process Creation Alert Type

Account Created and Deleted in Short Timeframe 5.5.0

Detects when an account was created and deleted in a short period of time.

This alert type has the following subtype category:

- Microsoft Entra Account Created and Deleted Within a Close Time Frame

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Valid Accounts (T1078)
- Tags: [Microsoft Entra; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `account_created_deleted_in_short_timeframe`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- correlation_info.0.targetResources.id, from the record in wineventlog representing a deletion
    - azure_ad.activityDisplayName: Delete user

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `user.name` — user name of the account involved in the activity
- `activityDisplayName` — name of the activity
- `category` — category of the activity
- `result` — result of the activity
- `resultReason` — result reason of the activity

Use Case with Data Points

This alert is triggered if an account (`user.name`) is created and then deleted within 24 hours. The Interflow includes the user account (`user.name`), the account creation time (`correlation_info[1].timestamp`), and the account deletion time (`correlation_info[0].timestamp`).

Alert Subtype: Microsoft Entra Account Created and Deleted Within a Close Time Frame

The Microsoft Entra Account Created and Deleted Within a Close Time Frame subtype is the same as the Account Created and Deleted in Short Timeframe alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `azure_ad_account_created_deleted`.

Account Creation Anomaly 5.4.0

An anomalously large number of accounts were created in a 5-minute interval or a user created an account for the first time in at least 14 days. Check the originating account and the new accounts for malicious behavior.

The data sources for this alert type are AWS, Microsoft Entra ID (formerly Azure AD), Google Workspace, and Windows.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [Internal] Persistence (TA0003)
- Technique: Create Account (T1136)
- Tags: [Internal; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `account_creation_anomaly`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_data.SubjectUserName
- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — SID of account creating new accounts

  or

- `event_data.SubjectUserName` — User name of account creating new accounts (for Windows only)

  The key field for this alert type can be `srcip_usersid` or `event_data.SubjectUserName`, depending on the data source.

- `srcip_username` — user name of account creating new accounts
- `responseElements.user.meta.createdBy` — AWS user creating new accounts
- `event_id` — Windows event ID corresponding to the account creations
- `actual` — Actual number of account creation events in a 5-minute period
- `typical` — Typical number of account creation events in a 5-minute period
- `event_summary.account_creation.new_users_count` — Count of user names of new accounts

- `event_summary.account_creation.new_users_summary` — Summary of user names of new accounts, which is a list of up to 100 unique user names associated with the new account creations

Use Case with Data Points

If an unusually high number of accounts were created (`actual`) by (`srcip_usersid`) in a 5-minute period, an alert is triggered. The Interflow record includes the data source, the account responsible for creating other accounts, the timestamp, the number of accounts created (`event_summary.account_creation.new_users_count`), and a list of the created accounts (`event_summary.account_creation.new_users_summary`).

Validation / Remediation

Investigate the account that is creating other accounts for a breach and investigate any activity from the newly created accounts.

Potential False Positives

False positives may occur when many accounts are created at the same time for business purposes, for example, when onboarding new users or otherwise procedurally generating accounts.

Account MFA Login Failure Anomaly

An anomalously large number of Multi-Factor Authentication (MFA) user login failures was observed for an account. Check with the user.

This alert type has the following subtypes:

- Machine Learning Anomaly
- Rule Based

Alert Subtype: Machine Learning Anomaly

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `cloud_account_login_failure_okta`.

Severity

45

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — cloud account user ID
- `srcip_username` — cloud account user name
- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `srcip_host` — host name of corresponding source IP address
- `login_type` — type of login
- `srcip_reputation` — source reputation
- `userIdentity.userName` — user name of the account involved in the event
- `eventSource` — source of the event
- `eventName` — name of the event
- `eventType` — type of the event

Use Case with Data Points

Multi-Factor Authentication login failures and successes are calculated periodically for every account (`srcip_usersid`). If the number of failures is significantly larger than the number of successes, an alert is triggered. A sample Interflow includes the login type (`login_type`), source host (srcip_host), and source reputation (`srcip_reputation`).

Alert Subtype: Rule Based

The Suspicious AWS Login Failure rules are used to identify suspicious AWS account login failures. Any one or more of these will trigger the AWS Cloud Account Login Failure alert type.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- eventName
- eventSource
- service_id
- stellar.rule_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `eventSource` — source of event
- `eventName` — name of event
- `eventType` — type of event

- `userIdentity.accountId` — key ID for the account involved in the event
- `userIdentity.userName` — user name of the account involved in the event
- `userIdentity.type` — type of account involved in the event
- `stellar.rule_id` — Stellar Cyber rule ID

Link to Rule-Based Alert Types

Rules Contributing to Suspicious AWS Login Failure

Application Usage Anomaly

An internal application had an anomalously large number of connections to one or more external hosts in a measured interval, exceeding 99.99% of all other intervals corresponding to different applications in the past two weeks. Investigate the application and connections, and consider blocking connections from the application.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR App Anomaly (XT2003)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `pripub_appid`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- appid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `appid` — application ID
- `appid_name` — application name
- `actual` — actual number of connections in the period
- `stellar.threshold` — threshold number of connections per interval below which 99.99% of all other intervals, corresponding to different applications in the past two weeks, fall
- `srcip_host` — host name of a sample source IP address
- `srcip_geo.countryName` — source country
- `dstip_host` — host name of a sample destination IP address

Use Case with Data Points

Every application's (`appid`) number of connections is calculated periodically. If an application's connections (`actual`) are larger than the threshold (`stellar.threshold`) below which 99.99% of all other intervals corresponding to different applications in the past two weeks fall, an alert is triggered. The Interflow includes a sample source host (`srcip_host`), the source country (`srcip_geo.countryName`), and a sample destination host (`dstip_host`). If there are multiple source or destination hosts, view the list in the Original Records.

AWS AMI Made Public

An AWS AMI was made public. Check with the user to make sure this was intentional.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: Privilege Escalation (TA0004)
- Technique: Valid Accounts (T1078)
- Tags: [AWS AMI; Access Control]

XDR Event Name

The `xdr_event.name` for this alert type in the Interflow data is `aws_ami_public`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- userIdentity.accountId

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `userIdentity.accountId` — key ID for the account
- `userIdentity.userName` — AWS account user name
- `userIdentity.type` — AWS account type
- `eventName` — AWS event name
- `eventSource` — AWS event source
- `eventType` — AWS event type

Use Case with Data Points

For each AWS account (`userIdentity.accountId`), activity to make an AMI public is monitored. If an AMI is made public, an alert is triggered. The Interflow includes the account ID (`userIdentity.accountId`), user name (`userIdentity.userName`), account type (`userIdentity.type`), AWS event name (`eventName`), AWS event source (`eventSource`), and AWS event type (`eventType`).

AWS Logging Stopped

AWS CloudTrail logging was stopped. Check with the user to make sure this was intentional.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Impair Defenses (T1562)
- Sub-technique: Disable or Modify Cloud Logs (T1562.008)
- Tags: [AWS CloudTrail]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `aws_stoplogging`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `userIdentity.accountId` — key ID for the account
- `userIdentity.userName` — AWS account user name
- `userIdentity.type` — AWS account type
- `eventName` — AWS event name
- `eventSource` — AWS event source
- `eventType` — AWS event type

Use Case with Data Points

For each AWS account (`userIdentity.accountId`), log disabling is monitored. Logging is enabled by default, so if logging is disabled, an alert is triggered. The Interflow includes the account ID (`userIdentity.accountId`), AWS account user name (`userIdentity.userName`), AWS account type (`userIdentity.type`), AWS event name (`eventName`), AWS event source (`eventSource`), and AWS event type (`eventType`).

AWS S3 Ransomware

Possible AWS S3 ransomware was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Impact (TA0040)
- Technique: Data Encrypted for Impact (T1486)
- Tags: [Malware; Ransomware; AWS S3]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `aws_s3_ransomware`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `userIdentity.accountId` — key ID for the account
- `userIdentity.userName` — AWS account user name
- `userIdentity.type` — AWS account type
- `eventName` — AWS event name
- `eventSource` — AWS event source
- `eventType` — AWS event type

Use Case with Data Points

For each AWS account user name (`userIdentity.userName`), suspicious S3 ransomware is monitored. If ransomware is detected, an alert is triggered. The Interflow includes the account ID (`userIdentity.accountId`), AWS account user name (`userIdentity.userName`), AWS account type (`userIdentity.type`), AWS event name (`eventName`), AWS event source (`eventSource`), and AWS event type (`eventType`).

Backup Catalogs Deleted by Ransomware

The `wbadmin.exe` utility was used to delete the backup catalog. Ransomware and other malware do this to prevent system recovery. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Indicator Removal on Host (T1070)
- Sub-technique: File Deletion (T1070.004)
- Tags: [Malware; Ransomware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ransomware_delete_backup_catalogs`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name 

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host on which the ransomware action happened
- `hostip_host` — host name
- `process_name` — name of the executed process
- `event_data.CommandLine` — command line that is executed to delete the backup catalog

Use Case with Data Points

If `wbadmin.exe` is used to delete the backup catalog, an alert is triggered. The Interflow includes the host IP address (`hostip`), process name (`process_name`), and command line (`event_data.CommandLine`).

Bad Destination Reputation Anomaly

A destination IP address with a bad reputation has received an anomalously large number of connections. Investigate the connections and consider blocking the destination IP address.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR Intel (XTA0005)
- Technique: XDR Bad Reputation (XT2010)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `dstip_bad_reps`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — destination IP address
- `dstip_host` — destination host name
- `dstip_reputation` — destination reputation
- `actual` — actual number of connections to the destination IP address in the period

- `typical` — typical number of connections to the destination IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `appid_name` — application name

Use Case with Data Points

The number of connections for every destination IP address (`dstip`) with a bad reputation (`dstip_reputation`) is calculated periodically. If a destination IP address's number of connections (`actual`) is much larger than the typical historical number (`typical`), an alert is triggered. The Interflow includes the source IP address making the connection (`srcip_host`), the application (`appid_name`) used, and the reputation of the source host (`srcip_reputation`).

Bad Reputation Login

A successful login was observed from an IP address with a history of malicious activity. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Bad Reputation (XT2010)
- Tags: [External]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `bad_reputation_login`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation (if not empty)
- `source_geo.countryName` — source country
- `dstip_host` — destination host name
- `login_type` — type of login
- `username` — user name

Use Case with Data Points

The login records are checked for every source IP address (`srcip`). If a source IP address has successful login records and its reputation (`srcip_reputation`) is bad (except brute-forcer and scanner), an alert is triggered. An example Interflow includes source IP address (`srcip`), source host (`srcip_host`), source reputation (`srcip_reputation`), source country (`srcip_geo.countryName`), login type (`login_type`), and user name (`username`).

Bad Source Reputation Anomaly

A source IP address with a bad reputation has made an anomalously large number of connections. Investigate the connections and consider blocking the source IP address.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Bad Reputation (XT2010)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `srcip_bad_reps`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `srcip_reputation` — source reputation
- `actual` — actual number of connections in the period
- `typical` — typical number of connections from the source IP address
- `dstip_host` — host name of corresponding destination IP address
- `dstip_reputation` — destination reputation
- `appid_name` — application name

Use Case with Data Points

The number of connections for every source IP address (`srcip`) with a bad reputation (`srcip_reputation`) is calculated periodically. If a source IP address's number of connections (`actual`) is much larger than the typical historical number (`typical`), an alert is triggered. The Interflow includes the application (`appid_name`) used and the reputation of the destination host (`dstip_reputation`).

Carbon Black: XDR Anomaly

The Carbon Black endpoint generates an anomalously high amount of log data or a rarely seen type of log data on the host. Investigate the device and the user, to see if this is expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR Anomaly (XT1000)
- Tags: [Carbon Black]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `carbonblack_edr_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- process.command_id
- host.ip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `host.ip` — device internal IP address
- `host.external_ip` — device external IP address
- `actual` — actual volume of log records in the period
- `typical` — typical difference in volume of log records between this period and the previous period

Use Case with Data Points

The number of occurrences of Carbon Black endpoint (cloud) log, based on the “UNKNOWN“ threat category (`event.type`), is tabulated periodically. If this category occurs (`actual`) much more often compared to its history (`typical`) or a rarely seen type of record is observed, an alert is triggered. The Interflow includes information such as the file name (`file.name`), process (`process.name`), and description (`xdr_event.description`).

Cloud Drive Data Exfiltration Anomaly 5.5.0

An account downloaded an atypical number of files over a 24-hour period. Check the user and the targeted files.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Exfiltration (TA0010)

- Technique: Exfiltration Over Web Service (T1567)
- Sub-technique: Exfiltration to Cloud Storage (T1567.002)
- Tags: [External; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `cloud_drive_data_exfiltration_anomaly`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_username

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_username` — user downloading an anomalous number of files
- `actual` — number of downloads in the last 24 hours
- `typical` — typical number of downloads in the last 24 hours
- `event_summary.cloud_drive_data_exfiltration_anomaly.downloaded_files_summary` — names of the downloaded files from Google Workspace, OneDrive, and SharePoint (on a per-user basis)

Use Case with Data Points

If an unusually high number of files were downloaded from a cloud service by (`srcip_username`) in a 24-hour period, an alert will be triggered. A sample Interflow record includes the user requesting downloads (`srcip_username`) and a summary of what files have been downloaded (`event_summary.cloud_drive_data_exfiltration_anomaly.downloaded_files_summary`).

Command & Control Reputation Anomaly

An anomalously large number of connections were made to known command and control servers. Investigate the connections and source hosts. If malicious, block the IP addresses of the command and control servers.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR Intel (XTA0005)
- Technique: XDR Command and Control Reputation (XT5001)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `cnc_reputation`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — destination IP address
- `dstip_host` — destination host name
- `dstip_reputation` — destination reputation
- `actual` — actual number of connections in the period
- `typical` — typical number of connections to the destination IP address with a C&C reputation
- `srcip_host` — host name of corresponding source IP address
- `srcip_reputation` — source reputation
- `appid_name` — application name

Use Case with Data Points

The number of connections for every destination IP (`dstip`) with a command and control reputation (`dstip_reputation`) is calculated periodically. If a destination IP has a much higher number of connections (`actual`) than its history (`typical`) in any period, an alert is triggered. The Interflow includes the application used in the connection (`appid_name`), the source host (`srcip_host`), and the source reputation (`srcip_reputation`).

Command Anomaly

A command has been executed an anomalously large number of times compared to its typical executions or those of other commands. Investigate the command and the user to determine if this is expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Execution (TA0002)
- Technique: Command and Scripting Interpreter (T1059)
- Tags: [Internal]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `command_anomaly`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- command 

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `command` — command executed
- `actual` — actual number of executions in the period
- `typical` — typical number of executions in the period
- `cwd` — current working directory from which the command executed
- `hostip` — host from which the command was run
- `hostip_host` — host name
- `username` — user name who ran the command

Use Case with Data Points

The number of times a command (`command`) has been executed is calculated periodically. If the volume (`actual`) is much larger than the typical volume (`typical`) of the command or other commands in any period, an alert is triggered. The Interflow includes the directory from which the command was executed (`cwd`), the host and source IP addresses (`hostip` and `srcip`) from which the command was executed, and the name of the user who ran the command (`username`).

Cryptojacking

An unauthorized coin miner used a computer to mine cryptocurrency. Consider blocking the source IP address.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Impact (TA0040)
- Technique: Resource Hijacking (T1496)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `cryptojacking`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `srcip` — source IP address of the cryptojacking action
- `dstip` — destination IP address of the cryptojacking action
- `srcip_reputation` — source reputation
- `srcip_host` — source host name
- `dstip_reputation` — destination reputation
- `dstip_host` — destination host name

Use Case with Data Points

If an unauthorized coin miner is detected by IDS, an alert is triggered. A sample Interflow includes the IDS signature (`ids.signature`), source IP address (`srcip`), source reputation (`srcip_reputation`), source host (`srcip_host`), destination IP address (`dstip`), destination reputation (`dstip_reputation`), and destination host (`dstip_host`).

CylanceOPTICS: XDR Anomaly

The Cylance OPTICS endpoint generates an anomalously high amount of log data or a rarely seen type of log data on the host. Investigate the device and the user to see if this is expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR Anomaly (XT1000)
- Tags: [Cylance]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `cylance_edr_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- tenantid
- event.description
- host.id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `host.name` — device name
- `event.description` — description of the detection rule
- `actual` — actual number of log records in the period

- `typical` — typical number of log records generated on the device

Use Case with Data Points

The number of occurrences of CylanceOPTICS log records (`event.provider`) is calculated periodically. If this category occurs (`actual`) much more often compared to its history (`typical`) or a rarely seen type of event is generated, an alert is triggered. The Interflow includes information such as the process name (`process.name`), parent process name (`process.parent.name`), and description (`event.description`).

Data Ingestion Volume Anomaly

A sensor is sending an anomalously high or low volume of data, compared to its typical volume. Check the sensor. A low volume could indicate a sensor failure or other problems. For a high volume, determine the cause of the increase.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: XDR SBA (XTA0003)
- Technique: XDR Bytes Anomaly (XT3001)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ade_outbytes_anomaly`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- engid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `engid` — sensor ID
- `engid_name` — sensor name
- `actual` — actual volume of data in the period
- `typical` — typical difference in data volume between this period and the previous period

Use Case with Data Points

The data ingestion volume of every data sensor with sensor id (`engid`) and sensor name (`engid_name`) is calculated periodically. If one of the following conditions is met, the anomaly is triggered:

- A moving window is used to record data ingestion volume. If the time window can be divided into two sub windows and the metric values of these two sub windows show

large deviation

- The ingestion volume is anomalously high compared to its own history
- The ingestion volume is anomalously low compared to its history and it keeps being low for a relatively longer period

A sample Interflow includes the sensor ID (`engid`) and sensor name (`engid_name`).

DGA

A host is using a potential Domain Generation Algorithm (DGA). If the target domain is a malicious domain, the host might be compromised. Investigate the DGA domains and the host.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Command and Control (TA0011)
- Technique: Dynamic Resolution (T1568)
- Sub-technique: Domain Generation Algorithms (T1568.002)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `dga_resolvable`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- metadata.request.effective_tld
- metadata.response.resolved_ips

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the host that sends the DGA queries
- `srcip_host` — source host name
- `metadata.request.effective_tld` — effective top-level domain of the DNS query
- `is_dga` — flag marking whether or not the DNS query is a DGA query
- `actual` — number of DGA domains the host has queried

Use Case with Data Points

Whenever a host (`srcip`) sends a DNS query (`appid_name`: `dns`) and the DNS server returns a non-existent domain (NXDOMAIN) response (`metadata.response.reply_code`), the NX domain query counter for the host is increased. We reset the counter if no NX domain queries are observed for a period of time. When the counter reaches a certain threshold, the host is

monitored. When monitored, we run the FQDNs of all DNS queries (`metadata.response.query`) sent by this host through domain generation analytics to determine whether the domain's entropy indicates a DGA anomaly. If so, we mark the DNS record (`is_dga`). If the DNS query gets a response with valid resolved IP addresses (`metadata.response.resolved_ips`), we call it a resolvable query, otherwise we call it a non-resolvable query.

If a monitored host (`srcip`) sends a resolvable DGA query (`is_dga`: `yes_resolvable`), we check the effective top-level domain (`metadata.response.effective_tld`). If the same host (`srcip`) previously sent non-resolvable DGA queries (`is_dga`: `yes`) with the same effective top-level domain (`metadata.response.effective_tld`), the host is considered to have a high risk of being compromised and performing C&C with DGA. The Interflow includes the source host (`srcip`), DNS query (`metadata.response.query`), query effective top-level domain (`metadata.response.effective_tld`), and DGA flag (`is_dga`).

DHCP Server Anomaly

A new DHCP server appeared in the network. This could be a hacker attempting to steer traffic. Investigate and consider telling employees to avoid this server.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Server Anomaly (XT2007)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `dhcp_anomaly`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- metadata.response.server_ip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.response.server_ip` — IP address of the anomalous DHCP server
- `dstip` — IP address of the anomalous DHCP destination
- `engid` — sensor that reported the DHCP traffic
- `srcip_host` — host name that visited the suspicious DHCP server
- `srcip_geo.countryName` — country name of the source that visited the suspicious DHCP server

Use Case with Data Points

If a DHCP server that has never been seen before appears in the network, an alert is triggered. The Interflow includes the destination IP address (`dstip`), destination host (`dstip_host`), source host (`srcip_host`), and source country (`srcip_geo.countryName`).

DNS Tunneling Anomaly

An anomalously large number of connections tunneling high-entropy traffic through DNS were made. This can indicate data exfiltration. Investigate the tunnel and source host. If malicious, block the source host.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Exfiltration (TA0010)
- Technique: Exfiltration Over Alternative Protocol (T1048)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `dns_tunnel`.

Severity

98

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.effective_tld
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `metadata.request.effective_tld` — effective top-level domain, such as yahoo.com
- `metadata.request.query` — DNS query
- `actual` — actual number of bytes transmitted through the tunnel in the period
- `typical` — typical number of bytes transmitted through a tunnel in the period
- `total_entropy` — total entropy (information density) sent by the DNS tunnel
- `query_count` — number of queries sent by the DNS tunnel

Use Case with Data Points

The DNS queries (`metadata.requests.query`) for each DNS tunnel (comprising the source host (`srcip_host`), destination host (`dstip`), and top-level domain (`effective_tld`)) are analyzed periodically. If a DNS tunnel has sent anomalously more entropy (`total_entropy`) and

bytes (`actual`) than is normal (`typical`) in any period, an alert is triggered. The number of queries sent (`query_count`) is also considered.

Dormant Account Anomaly 6.0.0

This topic covers a feature that is not available for all customers yet. See Early Access Program Features and Topics Under Development.

A user account has not logged any interactive actions such as logins, password changes, or resource accesses recently. Check the user account and consider disabling it.

Note the following:

- The data sources for the interactive actions are Active Directory, Microsoft Entra ID (formerly Azure AD), and Office365.
- For a user account to be eligible for dormant account alerts, sufficient historical interactive activity must be present to establish a baseline of prior activity.
- Reminder alerts are sent only if the user has a long history. There can be up to 3 alerts per user (at 14, 21, and 28 days), after which there will be no further alerts unless the user stops being dormant.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] Persistence (TA0003)
- Technique: Valid Accounts (T1078)
- Tags: [Identity Detection]

XDR Event Name

The `xdr_event.name` for this alert type in the Interflow data is `dormant_account`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_username

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_username` — source user name
- `days_silent` — number of days since last activity
- `last_seen_ts` — timestamp when the user last logged interactive data

Use Case with Data Points

If an account is observed to be dormant, an alert is triggered. The Interflow will contain a user name (`srcip_username`), the number of days since an interactive action was logged (`days_silent`), and the last seen timestamp (`last_seen_ts`) for that user.

Validation / Remediation

Analysts can validate the alert by checking whether activity is expected from that user. They can disable the account if it is not supposed to be dormant.

Potential False Positives

A false positive could be generated if some user activity is not counted as interactive.

Emerging Threat

An emerging threat has been observed. Investigate the IP address, domain name, URL, or file hashes (MD5, SHA1, SHA256) 5.4.0 and consider blocking.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR Intel (XTA0005)
- Technique: XDR Emerging Threat (XT5003)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `emerging_threat`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- domain_list
- url_list
- file.threat_indicator.file.hash.md5
- event_summary.process_list_hash.md5
- event_summary.file_list_hash.md5
- process.parent.threat_indicator.file.hash.md5
- process.threat_indicator.file.hash.md5
- file.threat_indicator.file.hash.sha1
- event_summary.process_list_hash.sha1
- event_summary.file_list_hash.sha1

- process.parent.threat_indicator.file.hash.sha1
- process.threat_indicat  e.hash.sha1
- file.threat_indicator.file.hash.sha256
- event_summary.process_list_hash.sha256
- event_summary.file_list_hash.sha256
- process.parent.threat_indicator.file.hash.sha256
- process.threat_indicator.file.hash.sha256

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address marked as an emerging threat
- `srcip_host` — host name of corresponding source IP address
- `dstip` — destination IP address marked as an emerging threat
- `dstip_host` — host name of corresponding destination IP address
- `url_list` — URL marked as an emerging threat
- `domain_list` — domain marked as an emerging threat
- `event_summary.detected_hashes.md5` — MD5 hash marked as emerging threat
- `event_summary.detected_hashes.sha1` — SHA1 hash marked as emerging threat
- `event_summary.detected_hashes.sha256` — SHA256 hash marked as emerging threat
- `event_summary.detected_file_names` — detected file names with matched hash value
- `event_summary.detected_process_names` — detected process names with matched hash value
- `event_summary.ioc_source_repu_summary.url_reputation` — URL reputation
- `event_summary.ioc_source_repu_summary.srcip_reputation` — source reputation
- `event_summary.ioc_source_repu_summary.dstip_reputation` — destination reputation
- `event_summary.ioc_source_repu_summary.domain_reputation` — domain reputation
- `event_summary.ioc_source_repu_summary.filehash_reputation` — hash reputation

Use Case with Data Points

Stellar Cyber monitors traffic for emerging threats. This alert type has file hash detection that monitors traffic, Syslog, and Windows events.

An alert is triggered if emerging threats are observed in any of the following:

- Source IP address (`srcip`)
- Destination IP address (`dstip`)
- URL (`url_list`)
- Domain (`domain_list`)

- MD5 file hash (`event_summary.detected_hashes.md5`)
- SHA1 file hash (`event_summary.detected_hashes.sha1`)
- SHA256 file hash (`event_summary.detected_hashes.sha256`)

Note that only one of these is needed to trigger the alert. So, although the Interflow includes the source IP address (`srcip`), destination IP address (`dstip`), URL (`url_list`), domain (`domain_list`), and file hashes (`event_summary.detected_hashes.md5`, `event_summary.detected_hashes.sha1`, and `event_summary.detected_hashes.256`), not all the values may be populated, depending on the nature of the observed threat.

Encoded PowerShell

A Windows host executed an encoded PowerShell script. Investigate the script contents to see if it is malicious. If so, consider quarantining the host.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Execution (TA0002)
- Technique: Command and Scripting Interpreter (T1059)
- Tags: [PowerShell]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `encoded_powershell`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the Windows host
- `hostip_host` — host name
- `event_data.ContextInfo` — PowerShell script context
- `event_data.Payload` — PowerShell script payload

Use Case with Data Points

If a Windows host (`srcip`) executes a PowerShell script whose context (`event_data.ContextInfo`) includes flags that indicate encoding or obfuscation of the script, an alert is triggered. The Interflow includes the IP address of the Windows host (`srcip`), the script context (`event_data.ContextInfo`), and script payload (`event_data.Payload`).

Rule Based Detection

The Encoded PowerShell rules are used to identify when a Windows host executed an encoded PowerShell script. Any one or more of these will trigger the Encoded PowerShell alert type.

The rule-based detection is the same as the Encoded PowerShell alert type above, with the following difference:

- The `detection_flag` is a Stellar Cyber enriched field:
  - **2100:** Encoded PowerShell
  - **2101:** Encoded PowerShell with hidden flag

Link to Rule-Based Alert Types

Rules Contributing to Encoded PowerShell Alert Type

Encrypted C&C

A connection to or from known command and control servers was observed in encrypted traffic. Consider blocking the source IP address.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Command and Control (TA0011)
- Technique: Encrypted Channel (T1573)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ssl_certificate`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the connection
- `dstip` — destination IP address of the connection
- `srcip_host` — host name of corresponding source IP address
- `srcip_geo.countryName` — source country of the connection
- `dstip_host` — host name of corresponding destination IP address
- `dstip_geo.countryName` — destination country of the connection

Use Case with Data Points

If known command and control servers are detected on either side of a connection with encrypted traffic, an alert is triggered. The Interflow includes the source IP address (`srcip`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination IP address (`dstip`), destination host (`dstip_host`), and destination country (`dstip_geo.countryName`).

Exploited C&C Connection

An exploited host with vulnerabilities initiated a connection to the exploit attacker, which could indicate the host being compromised and performing C&C activities. See if the exploit was successful. Check the source host, and consider blocking.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Command and Control Connection Exploitation (XT2014)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `exploit_attempt_correlation`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- exploit_attempt record's srcip, dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `tenant_id` — tenant ID
- `exploit_id` — ID of the original exploit event
- `seen_traffic_id` — ID of the original Interflow traffic record
- `srcip` (of exploit event) — IP address of the attacker (`correlation_info.srcip`)
- `dstip` (of exploit event) — IP address of the target host (`correlation_info.dstip`)
- `srcip` (of traffic record) — IP address of the target host (`correlation_info.srcip`)
- `dstip` (of traffic record) — IP address of the attacker (`correlation_info.dstip`)

Use Case with Data Points

Two events are involved in this alert type. In the first event, an attacker (`srcip`) with the IP address A is performing an exploit against a target (`dstip`) with the IP address B. If, following that event, an Interflow traffic record is observed where the target host (`srcip`) with IP address B initiates a network connection to the attacker (`dstip`) whose IP address is A, an alert is triggered.

When an alert is triggered a new correlation event is generated. The Interflow of the correlation event includes the reference ID of the exploit event (`exploit_id`), the reference ID of the traffic record (`seen_traffic_id`), the IP address of the attacker (`correlation_info.srcip` of the

exploit event or `correlation_info.dstip` of the traffic record), the IP address of the victim (`correlation_info.dstip` of the exploit event or `correlation_info.srcip` of the traffic record).

External Account Login Failure Anomaly 6.4.0

An anomalously large number of user login failures was observed for an account. Check with the user.

This alert type has the following subtypes:

- Office 365 / Entra ID
- Windows Security Events
- VPN Logon Events

This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay corresponding to their amount of latency.

The expected detection delay is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_cloud_account_login_failure`.

Severity

45

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — cloud account user ID

  or

- `srcip_username` — cloud account user name



The key field for this alert type can be either `srcip_usersid` or `srcip_username`, depending on the [redacted] feed.

- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `srcip_host` — host name of corresponding source IP address
- `login_type` — type of login
- `srcip_reputation` — source reputation

Use Case with Data Points

Login failures and successes are calculated periodically for every account (`srcip_usersid`). If the number of failures is significantly larger than the number of successes, an alert is triggered. A sample Interflow includes the login type (`login_type`), source host (`srcip_host`), and source reputation (`srcip_reputation`).

Alert Subtype: Office 365 / Entra ID 5.2.0

The Office 365 / Entra ID alert subtype is the same as the External Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Office 365 and Microsoft Entra ID (formerly Azure AD).
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_cloud_account_login_failure_o365_azure`.

Alert Subtype: Windows Security Events 5.2.0

The Windows Security Events alert subtype is the same as the External Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from all Windows security events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_cloud_account_login_failure_windows`.

Alert Subtype: VPN Logon Events 6.4.0

The VPN Logon Events alert subtype is the same as the External Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from all VPN logon events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_cloud_account_login_failure_vpn`.

The supported VPNs are FortiGate, Check Point, and Palo Alto Networks Panorama.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Brute-Forced Successful User Login

A successful login was observed from an IP address that had previously seen a large number of login failures, or a successful login to a user account was observed from another IP address or IP addresses that had previously seen a large number of login failures to that account. Check with the user.

This alert type has the following subtypes:

- Source IP Based
- User ID Based

This alert type has a relatively long detection delay of up to 40 minutes because it waits for login events from high latency data sources and is sensitive to the event order of user logins.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_user_success_brute_forcer`.

Severity

Alert Suppression Fields 

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- srcip_usersid

Alert Subtype: Source IP Based

The source IP-based alert subtype has the same **XDR Kill Chain** as the user ID-based alert subtype, but differs in the **Key Fields and Relevant Data Points** and **Use Case with Data Points**.

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_success_brute_forcer_srcip`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_usersid` — Windows SID associated with the source IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `source_geo.countryName` — source country
- `dstip_host` — destination host name
- `login_type` — type of login
- `username` — user name
- `related_alert._id` — link to the related External User Login Failure Anomaly

Use Case with Data Points

The login records are checked for every external source IP address (`srcip`). An alert is triggered if that IP address:

1. Has so many failed login attempts that it triggered the **External User Login Failure Anomaly**, and
2. Had a successful login

A sample Interflow includes the source IP address (`srcip`), login type (`login_type`), source host (`srcip_host`), source reputation (`srcip_reputation`), source country (`srcip_geo.countryName`), and user name (`username`).

Alert Subtype: User ID Based

The user ID-based alert subtype has the same **XDR Kill Chain** as the source IP-based alert subtype, but differs in the **Key Fields and Relevant Data Points** and **Use Case with Data Points**.

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_success_brute_forcer_srcip_usersid`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — Windows SID associated with the source IP address
- `srcip` — source IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `source_geo.countryName` — source country
- `dstip_host` — destination host name
- `login_type` — type of login
- `username` — user name
- `related_alert._id` — link to the related External Account Login Failure Anomaly

Use Case with Data Points

The login records to a user account (`srcip_usersid`) are checked for every external source IP address (`srcip`). An alert is triggered if that user account:

1. Has so many failed login attempts that it triggered the **External Account Login Failure Anomaly**, and
2. Had a successful login

A sample Interflow includes the source IP address (`srcip`), login type (`login_type`), source host (`srcip_host`), source reputation (`srcip_reputation`), source country (`srcip_geo.countryName`), and user name (`username`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Credential Stuffing

An anomalously large amount of username/password testing was observed on AWS, Okta, or Windows. Check the activity after successful logins, and consider blocking the source IP

addresses.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_credential_stuffing`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- msg_class
- service_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `msg_class` — name of the service: `cloudtrail` for AWS, `okta` for Okta, `Microsoft-Windows-Security-Auditing` for Windows
- `service_id` — specific account ID of a service
- `login_failure_rate` — rate of login failures per minute in the period
- `unknown_users_rate` — rate of unknown user names per minute in the period
- `unknown_users_to_login_failures` — ratio of unknown user names to login failures in the period
- `suspicious_ips` — suspicious source IP addresses (up to 100)
- `possible_breached_ips` — list of malicious IP addresses that may have successful breach activities

Use Case with Data Points

External credential stuffing is the constant testing of username/password combinations on the AWS, Okta, or Windows authentication functions. Login activity is monitored and if the number of failed logins is larger than normal for that service, an alert is triggered. The Interflow includes the service (`msg_class`), tenant's account ID on that service (`service_id`), suspicious source IP address (`suspicious_ips`), login failure rate (`login_failure_rate`), unknown user rate (`unknown_users_rate`), the ratio of unknown users to login failures (`unknown_users_to_login_failures`), and a list of source IP addresses that might have suspicious activities and should be investigated (`possible_breached_ips`).

More on Internal versus External and Inbound vs Outbound...

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External Exploited Vulnerability

A host with a vulnerability discovered by a security scanning tool was exploited by an attack on that same vulnerability, indicating a high probability of success. Check the target to see if it was compromised.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Exploited Vulnerability (XT2015)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_vuln_exploit_correlation`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- scan_record's srcip, dstip, ids.cve

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `tenantid` — tenant ID

- `vulnerability_id` — ID of the original security scan result
- `ids_event_id` — ID of the original IDS exploit event
- `srcip` (of security scan result) — IP address of the target `correlation_info.srcip`
- `dstip` (of IDS event) — IP address of the target (`correlation_info.dstip`)
- `srcip` (of IDS event) — IP address of the attacker (`correlation_info.srcip`)
- `vulnerability.cve` — CVE associated with the reported vulnerability
- `ids.cve` — CVE the attacker used to exploit the host

Use Case with Data Points

An attacker (`srcip`) with IP address A is performing an exploit against a target (`dstip`) with internal IP address B using a vulnerability (`ids.cve`) with CVE x. If any security scanning tool found the target (`srcip`) with IP address B to have a vulnerability (`vulnerability.cve`) with CVE x, an alert is triggered.

When an alert is triggered, a new correlation event is generated. The Interflow of the correlation event includes the ID of the IDS exploit event (`ids_event_id`), the ID of the security scan record (`vulnerability_id`), the IP address of the attacker (`correlation_info.srcip` of the IDS event), the IP address of the victim (`correlation_info.dstip` of the IDS event or `correlation_info.srcip` of the security scan record), and the CVE that was used in the exploit (`vulnerability.cve` and `ids.cve`).

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External Firewall Denial Anomaly

A source host had actions blocked by a firewall too many times. Investigate the firewall rules that were violated. If suspicious, block the source IP address.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts

- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Firewall Anomaly (XT2002)
- Tags: [External; Firewall Anomalies; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_fw_action`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_host

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source host IP address
- `srcip_host` — source host IP address
- `actual` — actual number of firewall denials in the period
- `typical` — typical number of firewall denials in the period
- `dstip_host` — host name of corresponding destination IP address
- `dev_name` — name of the firewall
- `engid_name` — name of the sensor

Use Case with Data Points

The number of firewall denials for every source IP address (`srcip`) is calculated periodically. If a source IP address's number of firewall denials (`actual`) is much larger than the historical count (`typical`) of all IP addresses, an alert is triggered. The Interflow includes the name of the firewall (`dev_name`), the name of the sensor (`engid_name`), and the destination host (`dstip_host`).

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External Firewall Policy Anomaly

A firewall policy was triggered that has never been seen before (or has very rarely been seen). Investigate the policy, the intent, and the associated traffic to determine whether this activity is legitimate or malicious.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Firewall Anomaly (XT2002)
- Tags: [External; Firewall Anomalies; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_fw_policy_id`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- fw_policy_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `fw_policy_id` — ID of the violated firewall policy
- `days_silent` — number of days since this firewall policy was last seen
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `dev_name` — device name
- `dev_type` — device type
- `engid_name` — sensor name

Use Case with Data Points

A firewall policy ID (`fw_policy_id`), which is raised by a device (`dev_name` and `dev_type`) and captured by a sensor (`engid_name`), shows never seen or very rare (`days_silent`) traffic between a host (`srcip_host`) and another host (`dstip_host`). This will trigger an alert.

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External Handshake Failure

There were too many handshake failures between two hosts, which might indicate port scanning. Check the source host to see if this was expected and, if not, consider blocking the host.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Reconnaissance (TA0043)
- Technique: Active Scanning (T1595)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_handshake_failure`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- appid_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the host with the handshake failures
- `srcip_host` — source host name
- `dstip` — destination IP address of the host with the handshake failures
- `dstip_host` — destination host name
- `timestamp` — when the scan happened

Use Case with Data Points

If a host (`srcip`) scans across many ports on another host (`dstip`), an alert is triggered. The Interflow includes the IP address of the potential attacker (`srcip`), the IP address of the victim (`dstip`), a special message flag (`msgtyp`), and when the scan happened (`timestamp`).

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External IDS Signature Spike

A source IP address transmitted an anomalous number of different IDS signatures. Typically, this indicates host penetration or vulnerability scanning.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_ids_signature_spike`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_summary.ids_signatures_summarize` — summarized IDS signatures of the exploit
- `srcip_host` — source host name
- `actual` — actual number of unique IDS signatures in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of unique IDS signatures from the source IP address, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1

Use Case with Data Points

The number of unique IDS signatures (`ids.signature`), weighted by their severity (`ids.severity`), are calculated periodically. If many different exploits with unique IDS signatures are observed, an alert is triggered. The Interflow includes a source (`srcip`), timestamp, an accumulated severity of IDS signatures (`actual`), the usual accumulated severity of IDS signatures (`typical`), and a sampling of the IDS signatures used in the attack (`ids_signatures_summarize`).

More on Internal versus External and Inbound vs Outbound...

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External IP / Port Scan Anomaly

A host has either generated an anomalous number of connections compared to the typical amount, or has triggered an anomalous number of connection failure responses, in the measured interval. This can indicate that an attacker is scanning for computers or ports to exploit. If the source IP address is internal targeting an external address, check with the user. If the source IP address is external targeting any addresses, it could be a scanning campaign.

This alert type has the following subtypes:

- Connection Failure Anomaly (Sensor Traffic)
- Connection Spike Anomaly (Firewall / Windows Traffic)

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Reconnaissance (TA0043)
- Technique: Active Scanning (T1595)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_port_scan`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Alert Subtype: Connection Failure Anomaly (Sensor Traffic)

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `connection_failure_anomaly`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `num_failed` — unique number of (destination IP and destination port) tuples that respond with failed status
- `num_successful` — unique number of (destination IP and destination port) tuples that respond with success status
- `percent_failed` — percent of unique (destination IP and destination port) tuples that respond with failed status
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.

- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name
- `summary_port_scan_type` — port scan type 6.2.0

Example of Event Summarization in the Alert Description

Within a 5.0-minute period the overall traffic from 10.1.1.1 consisted of the following scans: 64.94% to 10.2.1.26/31 on port(s) ranging from 2 to 49157, 35.06% to 10.2.1.28/30 on port(s) ranging from 1 to 49157.

Use Case with Data Points

For every unique triplet (source IP address, destination IP address, and destination port) browsed by each source IP address (`srcip`), the number of response failures and successes is calculated periodically. If the number of failures is significantly larger than the number of successes, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination host (`dstip_host`), and application name (`appid_name`).

Considering that a lateral scan (private to private) is more sensitive than a non-lateral scan, this alert type is divided into two parts. One focuses on lateral scan analysis, the other focuses on non-lateral scan analysis. The mechanism remains the same as before, with the trigger condition for lateral scan alert being more sensitive than non-lateral one.

Validation / Remediation

If the source IP address is internal targeting an external address, check with the user if they are aware of the activity or if they are authorized to perform the activity. Inform the user's supervisor if the activity is unauthorized.

If the source IP address is external targeting any addresses, check the reputation of the source IP address as in known malicious/scanner.

Potential False Positives

Some legitimate activities such as vulnerability scans or penetration testing may trigger this alert type, if from an external IP address to an internal IP address.

Alert Subtype: Connection Spike Anomaly (Firewall / Windows Traffic)

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `connection_spike_anomaly`.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Reconnaissance (TA0043)
- Technique: Active Scanning (T1595)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_port_scan_tsa`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of connections to the destination IP address in the time period
- `typical` — typical number of connections to the destination IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name
- `summary_port_scan_type` — port scan type 6.2.0

Example of Event Summarization in the Alert Description

Within a 5.0-minute period the overall traffic from 10.1.1.1 consisted of the following scans: 64.94% to 10.2.1.26/31 on port(s) ranging from 2 to 49157, 35.06% to 10.2.1.28/30 on port(s) ranging from 1 to 49157.

Use Case with Data Points

For every unique (destination IP address and destination port) browsed by each source IP address (`srcip`), the number of response failures and successes and the number of total data volume are calculated periodically. If the total data volume is significantly larger than the typical number, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination host (`dstip_host`), and application name (`appid_name`).

Considering that a lateral scan (private to private) is more sensitive than a non-lateral scan, this alert type is divided into two parts. One focuses on lateral scan analysis, the other focuses on non-lateral scan analysis. The mechanism remains the same as before, with the trigger condition for lateral scan alert being more sensitive than non-lateral one.

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If the source IP address is internal targeting an external address, check with the user if they are aware of the activity or if they are authorized to perform the activity. Inform the user's supervisor if the activity is unauthorized.

If the source IP address is external targeting any addresses, check the reputation of the source IP address as in known malicious/scanner.

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External Non-Standard Port Anomaly

An application had an anomalously large number of connections or a rarely seen connection on non-standard ports. Check the application to be sure this is benign.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] Command and Control (TA0011)
- Technique: Non-Standard Port (T1571)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_non_std_port_anomaly`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- dstport
- appid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — destination IP address
- `dstport` — destination port
- `appid` — application ID
- `days_silent` — number of days since the application was last seen
- `appid_name` — application name
- `dstip_host` — host name of corresponding destination IP address
- `actual` — actual number of connections in the period
- `typical` — typical number of connections in the period

Use Case with Data Points

The number of connections for an application (`dst_ip` + `dstport` + `appid`) is calculated periodically. If a non-standard combination has an actual number of connections (`actual`) that is much larger than the typical number of connections (`typical`), or the combination has not appeared for a long time, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination IP address (`dstip`), destination port (`dstport`), application ID (`appid`), and application name (`appid_name`).

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External Other Malware

Malware with uncategorized malicious activity was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] XDR Malware (XTA0006)

- Technique: XDR Miscellaneous Malware (XT6001)
- Tags: [External; Malwa

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_malware_activity`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `ids.severity` — severity of the IDS signature 5.2.0
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the malware
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates malware that cannot be categorized as ransomware, spyware, trojan, PUA, or adware, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the malware (`file_name`) from the sandbox.

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- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Password Spraying

.An anomalously large number of failed logins with unknown user names was observed on external Windows authentication services. Check the activity after successful logins, and consider blocking the source IP addresses.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Sub-technique: Password Spraying (T1110.003)
- Tags: [External; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_password_spray`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_data.Workstation
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address generating a failed login

or

- `event_data.Workstation` — workstation generating a failed login



The key field for this alert type can be either `srcip` or `event_data.Workstation`, depending on the data feed.

- `srcip_host` — source host name
- `event_id` — Windows event ID corresponding to the login failures
- `login_type` — type of login protocol; the available values vary by `event_id`
- `actual` — actual number of failed logins with unknown user names in a 5-minute period
- `typical` — typical number of failed logins with unknown user names in a 5-minute period
- `event_summary.password_spray_user_summary` — list of up to 100 unknown user names associated with the failed logins (the first three are shown in the alert description)

Use Case with Data Points

If a potential password spraying attack is observed, an alert is triggered. The Interflow includes a source (`srcip` or `event_data.Workstation`), timestamp, the type of login (`login_type`), the number of failed logins (`actual`), the usual number of failed logins (`typical`), and a sampling of the user names used in the attack (`password_spray_user_summary`).

More on Internal versus External and Inbound vs Outbound...

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- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External PII Leaked

Personally identifiable information (social security numbers or credit cards) has been observed in the clear. Check the source to see if it is compromised. If so, consider blocking it.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Exfiltration (TA0010)
- Technique: Automated Exfiltration (T1020)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_pii_leak`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the PII leak
- `dstip` — destination IP address of the PII leak
- `ids.signature` — IDS signature
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

If a personally identifiable information leak is detected by IDS, an alert is triggered. A sample Interflow includes the IDS signature (`ids.signature`), source IP address (`srcip`), destination IP address (`dstip`), source host (`srcip_host`), and destination host (`dstip_host`).

> *More on Internal versus External and Inbound vs Outbound...*
>
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list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Plain Text Passwords Detected

A plain text password was detected in unencrypted traffic. Check with the user.

This alert type looks for the presence of `metadata.request.password`, `metadata.request.auth_password`, and `metadata.request.auth_simple` in the Interflow records from the sensors. When plain text passwords are present in the network traffic, the sensors are able to decode and create the corresponding Interflow fields. To preserve privacy, the actual passwords are replaced by a sequence of asterisks (*).

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Clear Password (XT2006)
- Tags: [External; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_clear_password`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `actual` — actual number of connections with a plain text password in the period
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name

Use Case with Data Points

If there are plain text passwords in unencrypted traffic records with a public source IP address (`srcip`) or destination IP address (`dstip`), an alert is triggered. A sample Interflow includes the

source IP address (`srcip`), destination IP address (`dstip`), source host (`srcip_host`), destination host (`dstip_host`), and application (`appid_name`).

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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Protocol Account Login Failure Anomaly

An anomalously large number of login failures over SMB or FTP was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_protocol_account_login_failure`.

Severity

35

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.username

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.request.username` — user name in the HTTP connection request
- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `appid_name` — application name
- `login_type` — type of login
- `srcip_host` — host name of corresponding source IP address
- `srcip_reputation` — source reputation

Use Case with Data Points

For every user name (`metadata.request.username`) in the HTTP connections names (that do not begin with "Mozilla" or "Aella"), the number of failed and successful logins are calculated periodically. If the number of failed logins is much greater than successful logins, an alert is triggered. The Interflow includes the application name (`appid_name`), login type (`login_type`), source host (`srcip_host`), and source reputation (`srcip_reputation`).

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External PUA

Unwanted applications or malware that bombards the user with advertisements has been observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] XDR Malware (XTA0006)
- Technique: XDR PUA (XT6002)
- Tags: [External; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_pua`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the PUA
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates potentially unwanted applications (PUA), an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`) or IDS signature for ML-IDS (`ids.signature`), along with event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the PUA (`file_name`) from the sandbox.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

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- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Ransomware

Malware that prevents you from accessing your system or files and demands ransom payment in order to regain access was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Impact (TA0040)
- Technique: Data Encrypted for Impact (T1486)
- Tags: [External; Malware; Ransomware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_ransomware`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the ransomware
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates ransomware, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the ransomware (`file_name`) from the sandbox.

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External RDP BlueKeep

Use of a scanner by zerosum0x0 that discovers targets vulnerable to BlueKeep (CVE-2019-0708) has been observed. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [External] Privil Escalation (TA0004)
- Technique: Exploitation for Privilege Escalation (T1068)
- Tags: [External; RDP; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_rdp_bluekeep`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- ids.signature
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `srcip_host` — source host name
- `dstip_host` — destination host name

Use Case with Data Points

If the scanner by zerosum0x0 is used, an alert is triggered. A sample Interflow includes the IDS signature (`ids.signature`), source host (`srcip_host`), and destination host (`dstip_host`).

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INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External RDP Brute Force Attack

An anomalously large number of RDP connections to an RDP server was observed. Check the source IP addresses to determine whether they are unknown or malicious, and monitor any successful RDP logins.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; RDP; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_rdp_brute_force`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — IP address of the destination RDP server
- `dstip_host` — destination host name
- `actual` — actual number of RDP connections to the destination IP address in the observed time bucket
- `typical` — typical number of RDP connections to the destination IP address in most time buckets
- `srcip` — source IP address
- `srcip_host` — source host name

Use Case with Data Points

RDP connection activity is monitored and the number of connections are calculated periodically. If the number of connections to an RDP server (`actual`) is much greater than normal (`typical`), an alert is triggered. A sample Interflow includes the destination IP address (`dstip`) and source IP address (`srcip`).

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External RDP Suspicious Outbound

Non-standard tools connecting to TCP port 3389 were observed. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR App Anomaly (XT2003)
- Tags: [External; RDP; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_rdp_suspicious_outbound`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the host that connects to TCP port 3389 with a non-standard tool

- `srcip_host` — source host name
- `process_name` — process name

Use Case with Data Points

Connections to TCP port 3389 are monitored, and if non-standard tools connect, an alert is triggered. A sample Interflow includes the source IP address (`srcip`) and the process name (`process_name`). The following are the standard tools:

- mstsc.exe
- RTSApp.exe
- RTS2App.exe
- RDCMan.exe
- ws_TunnelService.exe
- RSSensor.exe
- RemoteDesktopManagerFree.exe
- RemoteDesktopManager.exe
- RemoteDesktopManager64.exe
- mRemoteNG.exe
- mRemote.exe
- Terminals.exe
- spiceworks-finder.exe
- FSDiscovery.exe
- FSAssessment.exe
- MobaRTE.exe
- chrome.exe
- thor.exe
- thor64.exe

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External Scanner Behavior Anomaly

An anomalously large amount of scanning behavior or a rarely seen scan behavior was found. Cross-check with the **IP / Port Scan Anomaly** alert.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Reconnaissance (TA0043)
- Technique: Active Scanning (T1595)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_scan_anomalies`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — signature of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name

Use Case with Data Points

The number of occurrences of each scanner, based on IDS signature (`ids.signature`), is calculated periodically. If one scanner occurs (`actual`) much more often than its history (`typical`), an alert is triggered. The Interflow includes information such as the traffic application type (`appid_name`), source (`srcip_host`), and destination (`dstip_host`).

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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SMB Read Anomaly

An IP address sent an anomalously large number of read requests to SMB protocol based service(s). Investigate the files that the IP address tried to read. If suspicious, block the source IP address.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; SMB; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_smb_read_anomaly`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — source host name
- `actual` — actual number of SMB reads from the source IP address in the period
- `typical` — typical number of SMB reads from other source IP addresses in the period
- `dstip_host` — destination host name

- `smb_username` — SMB user name
- `event_summary.smb_r[illegible]h_list` — folders experiencing a high volume of SMB read requests (the first three are shown in the alert description) 5.2.0

Use Case with Data Points

The number of SMB read requests for every source IP address (`srcip`) is calculated periodically. If a source IP address's number of SMB reads (`actual`) is much larger than the typical number (`typical`) and that of other IP addresses in any period, an alert is triggered. The Interflow includes the SMB user (`smb_username`) and destination host (`dstip_host`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SMB Username Enumeration

At least 5 different users SMB login attempts and 1 denied attempt or at least 10 different users SMB login attempts, were observed from the same source. Check the source IP address. If malicious, consider blocking it.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; SMB; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_smb_user_scan`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `actual` — actual unique SMB user count
- `typical` — SMB user count threshold
- `smb_username_set` — all SMB login user names

Use Case with Data Points

If one source IP address (`srcip`) has several SMB login attempts with (1) at least 5 unique user names and at least 1 denied attempt or (2) at least 10 unique user names, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), destination host (`dstip_host`), and all the user names (`smb_username_set`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SMB Write Anomaly

An IP address sent an anomalously large number of SMB write requests. Investigate the files that the IP address tried to write. If suspicious, block the source IP address.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Impact (TA0040)
- Technique: Data Manipulation (T1565)
- Tags: [External; SMB; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_smb_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — source host name
- `actual` — actual number of SMB writes in the period
- `typical` — typical number of SMB writes in the period
- `dstip_host` — destination host name
- `smb_username` — SMB user name
- `event_summary.smb_path_list` — folders experiencing a high volume of SMB write requests (the first three are shown in the alert description) 5.2.0

Use Case with Data Points

The number of SMB write requests for every source IP address (`srcip_host`) is calculated periodically. If a source IP address's number of SMB writes (`actual`) is much larger than the typical number (`typical`) and that of other IP addresses in any period, an alert is triggered. The Interflow includes the SMB user (`smb_username`) and destination host (`dstip_host`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.

- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Spyware

Malware that collects and shares information about a device without consent was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] XDR Malware (XTA0006)
- Technique: XDR Spyware (XT6003)
- Tags: [External; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_spyware_activity`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country

- `file_name` — name of the file that carries the spyware
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates spyware activity, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the spyware (`file_name`) from the sandbox.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SQL Anomaly

An IP address sent an anomalously large number of queries to one or more SQL servers. Investigate the queries. If suspicious, block the source IP address.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Database; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_mysql_anomaly`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — source host name
- `srcip_geo.countryName` — name of the source country
- `actual` — actual number of SQL queries in the period
- `typical` — typical number of SQL queries from the source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of SQL queries for every source IP address (`srcip_host`) is calculated periodically. If a source IP's SQL query count (`actual`) is much larger than the typical count (`typical`) and that of other IP addresses in any period, an alert is triggered. The source IP's country is (`srcip_geo.countryName`). The Interflow includes the destination host (`dstip_host`) the source IP visits.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SQL Dumpfile Execution

The SQL `dumpfile` command was observed. This command is commonly used to dump database content or query output to a file on disk. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exploitation
- Tactic: [External] Collection (TA0009)
- Technique: Data Staged (T1074)
- Tags: [External; Database; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_sql_db_dump`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `actual` — number of SQL `dumpfile` queries
- `srcip_host` — source host name
- `source_geo.countryName` — source country
- `dstip_host` — destination host name

Use Case with Data Points

If the SQL `dumpfile` command is seen on any source IP address (`srcip`), an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), and number of SQL `dumpfile` queries in the period (`actual`).

> *More on Internal versus External and Inbound vs Outbound...*
>
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>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.

- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SQL Shell Command

Shell commands were observed over a SQL connection, which is a common way hackers try to gain shell access over vulnerable SQL applications. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] Execution (TA0002)
- Technique: Command and Scripting Interpreter (T1059)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_database_command`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `dstip` — destination IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `dstip_host` — destination host name
- `dstip_reputation` — destination reputation
- `metadata.request.query` — SQL query command
- `actual` — number of query records from one source to one destination in one period

Use Case with Data Points

For SQL query records, if special commands (such as `select mylab_sys_exec`) are found, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), destination IP

address (`dstip`), source host (`srcip_host`), source reputation (`srcip_reputation`), destination host (`dstip_host`), destination reputation (`dstip_reputation`), and SQL query records (`metadata.request.query`).

More on Internal versus External and Inbound vs Outbound...

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- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Suspected Malicious User Agent

An external HTTP connection was made by a user agent that has been identified as potentially malicious. Investigate the connection's destination.

This alert type has the following subtypes:

- Predicted Malicious Agent
- Known Malicious Agent Match

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR User Agent Anomaly (XT2012)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_suspected_malicious_user_agent`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.user_agent

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `user_agent.original` — user agent in the HTTP connection request
- `metadata.request.user_agent` — user agent in the HTTP connection request
- `stellar.confidence` — model's confidence in the prediction
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name

Use Case with Data Points

If a seen user agent is identified as suspicious, an alert is triggered. The alert includes the suspicious user agent (`metadata.request.user_agent`), confidence (`stellar.confidence`), tenant (`tenant_name`), source IP (`srcip`), and destination IP (`dstip`) in the key fields. Additionally, the confidence level of the model is displayed in the alert description in a pop-up box.

Alert Subtype: Predicted Malicious Agent

The Predicted Malicious Agent alert subtype is the same as the External Suspected Malicious User Agent alert type above, with the following differences:

- The `stellar.anomaly_tag` is `predicted_external`.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_suspected_malicious_user_agent`.
- It is triggered by a machine learning classifier.

Alert Subtype: Known Malicious Agent Match

The Known Malicious Agent Match alert subtype is the same as the External Suspected Malicious User Agent alert type above, with the following differences:

- The `stellar.anomaly_tag` is `known_external`.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_suspected_malicious_user_agent_known_malicious`.
- It is triggered by known threats.

More on Internal versus External and Inbound vs Outbound...

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- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).

- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Suspected SQL Injection 6.4.0 6.6.0

A source host is exhibiting external SQL injection behavior with multiple queries observed in the last 60 minutes.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_http_sqli`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — source host name
- `dstip_host` — destination host name
- `dstport` — destination port
- `event_summary.sqli.top_sqli_uris` — suspected queries from source host observed in time window (top 5 shown when more than 5 are detected)
- `stellar.sqli_variation` — unique queries from the source host observed in the time window

- `stellar.sqli_duration` — time window for suspicious queries observation from source host

Use Case with Data Points

An external host is attacking your web services with SQL injection attacks using many unique payloads observed in the last 60 minutes. Consider blocking this source and check if the attack succeeds.

> *More on Internal versus External and Inbound vs Outbound...*
>
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External SYN Flood Attacker

An attacker sends a large amount of SYN requests to a target's system in an attempt to consume enough server resources to make the system unresponsive to legitimate traffic.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Impact (TA0040)
- Technique: Endpoint Denial of Service (T1499)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_syn_flood_attacker`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the SYN flood
- `srcip_host` — source host name
- `dstip` — target IP address of the SYN flood
- `dstip_host` — destination host name
- `dstport` — port on target host that received the SYN flood
- `syn_flood_events` — number of SYN packets during the period

Use Case with Data Points

If an external host (`srcip`) sends too many SYN packets (`syn_flood_events`) to internal target(s) (`dstip`) in a five-minute time window, an alert is triggered. The Interflow includes the IP address of the source host (`srcip`), the IP address of the target host (`dstip`), the port of the target host (`dstport`), and how many SYN packets were observed (`actual`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
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> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External SYN Flood Victim

A large amount of SYN requests were observed, which can indicate an attempt to consume server resources and make the target unresponsive. Check to see if the host is malicious or compromised. If so, consider blocking the source host.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Impact (TA0040)

- Technique: Endpoint Denial of Service (T1499)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_syn_flood`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the SYN flood
- `srcip_host` — source host name
- `dstip` — target IP address of the SYN flood
- `dstip_host` — destination host name
- `dstport` — port on target host that received the SYN flood
- `syn_flood_events` — number of SYN packets during the period

Use Case with Data Points

If an external host (`srcip`) sends too many SYN packets (`syn_flood_events`) to internal target(s) (`dstip`) in a five-minute time window, an alert is triggered. The Interflow includes the IP address of the source host (`srcip`), the IP address of the target host (`dstip`), the port of the target host (`dstport`), and how many SYN packets were observed (`actual`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction

of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External Trojan

Malware that disguises itself as legitimate software in order to gain access to a system or files has been observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] XDR Malware (XTA0006)
- Technique: XDR Trojan (XT6004)
- Tags: [External; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_trojan_activity`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the trojan
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates trojan activity, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for

ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the trojan (`file_name`) from the sandbox.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External URL Reconnaissance Anomaly

An anomalous number of HTTP 4xx errors were observed. This can indicate an attacker scanning for pages to exploit. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Reconnaissance (TA0043)
- Technique: Active Scanning (T1595)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_url_scan`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `event_summary.total_failed` — number of unique URLs with HTTP error status response in the period
- `event_summary.total_successful` — number of unique URLs with HTTP success status response in the period
- `event_summary.total_fail_ratio` — percent of unique URLs with HTTP error status response in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `srcip_geo.countryName` — source country name

Use Case with Data Points

For every unique URL browsed by each source IP address (`srcip`), the number of HTTP response failures and successes is calculated periodically. If the number of failures is significantly larger than the number of successes, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination host (`dstip_host`), and source country (`srcip_geo.countryName`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External User Application Usage Anomaly

A user who typically uses a small, consistent number of applications used a new application. Investigate the application, to see if it is benign. Check with the user to see if this was expected.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR App Anomaly (XT2003)
- Tags: [External; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_user_uncommon_app`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — source user ID
- `appid_name` — application name
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_family` — application family
- `srcip_username` — source user name
- `stability` — score measuring the time since the last new application was used
- `days_stable` — time since the last new application was used
- `diversity` — score measuring the number of applications that the user used
- `child_count` — number of applications that the user used

Use Case with Data Points

An alert is triggered under the following conditions:

- a user (`srcip_usersid`, `srcip_username`) with a small number of applications (`diversity`, `child_count`) who has not used a new application for a long period of time (`stability`, `days_stable`), and then
- a new application (`appid_name`) belonging to an application family (`appid_family`) appears on a host (`scrip_host`) with this user, and
- that host connects to another host (`scrip_host`)

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External User Data Volume Anomaly

A user had an anomalously large volume of traffic compared to its typical volume or that of its peers. Investigate the user to determine if this is expected.



Firewall and non-firewall data do not contribute to the same alert, so this alert will have either entirely firewall data or no firewall data.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Bytes Anomaly (XT3001)
- Tags: [External; User Behavior Analytics; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_user_bytes_sum`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — source user ID
- `actual` — actual traffic volume in the period
- `typical` — typical traffic volume from the user
- `srcip_host` — host name of corresponding source IP address
- `srcip_username` — source user name
- `dstip_host` — host name of corresponding destination IP address
- `dstip_reputation` — destination reputation
- `dstip_geo.countryName` — destination country
- `appid_name` — application name

Use Case with Data Points

The total traffic volume of each user identified by user ID (`scrip_usersid`) is calculated periodically. If the volume in one period (`actual`) is much larger than its normal volume (`typical`), an alert is triggered.

The Interflow includes the source IP address (`scrip_host`), destination IP address (`dstip_host`), destination reputation (`dstip_reputation`), destination country (`dstip_geo.countryName`), and application of the traffic (`appid_name`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

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- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

External User Login Failure Anomaly 6.4.0

An anomalous number of login failures was observed for one of the following applications: SSH, SMTP, FTP, RDP, SMB, databases, Active Directory, Office 365, Okta, AWS CloudTrail, VPNs, or Google Workspace. For Okta, an anomalous number of multi-factor authentication (MFA) failures was observed. Check with the user.

This alert type has the following subtypes:

- Office 365 / Entra ID

- Source IP Based
- Destination IP Based
- Kerberos Events
- Source IP Based Windows Logon Events
- Destination IP Based Windows Logon Events
- VPN Logon Events

This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay corresponding to their amount of latency.

The expected detection delay is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_user_login_fail`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `dstip` — destination IP address
- `dstip_host` — destination host name
- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold

will generate an alert.

- `login_type` — type of login, such as `ssh_traffic`, `okta_log`, or `aws_cloudtrail`
- `srcip_host` — source host name
- `srcip_reputation` — source reputation

Use Case with Data Points

Login failures and successes are calculated periodically for every source (`srcip`) and destination (`dstip`) IP address. If the number of failures is significantly larger than the number of successes, an alert is triggered. The Interflow includes the login type (`login_type`), source host (`srcip_host`), and source reputation (`srcip_reputation`).

Alert Subtype: Office 365 / Entra ID 5.2.0

The Office 365 / Entra ID alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Office 365 and Microsoft Entra ID (formerly Azure AD).
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_o365_azure`.

Alert Subtype: Source IP Based 5.2.0

The Source IP-based alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from network traffic, system logs, Linux events, and AWS events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_srcip`.

Alert Subtype: Destination IP Based 5.2.0

The Destination IP-based alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from network traffic, system logs, Linux events, and AWS events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_dstip`.

Alert Subtype: Kerberos Events 5.2.0

The Kerberos Events alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Kerberos events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_kerberos`.

Alert Subtype: Source IP Based Windows Logon Events 5.2.0

The Source IP-based Windows Logon Events alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Windows logon events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_src_win_logon`.

Alert Subtype: Destination IP Based Windows Logon Events 5.2.0

The Destination IP-based Windows Logon Events alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Windows logon events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_dst_win_logon`.

Alert Subtype: VPN Logon Events 6.4.0

The VPN Logon Events alert subtype is the same as the External User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from all VPN logon events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `external_user_login_fail_vpn`.

The supported VPNs are FortiGate, Check Point, and Palo Alto Networks Panorama.

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- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

File Action Anomaly

Actions, such as move, copy, delete, or change attribute, were taken on a file or files an anomalous number of times. Investigate the actions and the user to see if this is expected.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Impact (TA0040)
- Technique: Data Manipulation (T1565)
- Tags: [Internal; File Anomaly]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `anomalous_file_action`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- secondary

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `secondary` — user name
- `actual` — actual number of file actions in the period
- `typical` — typical number of file actions in the period
- `path` — path to the file

Use Case with Data Points

The number of file actions for each user (`secondary`) is calculated periodically. If the volume (`actual`) is anomalous compared to the typical volume (`typical`) of file actions in any period, an alert is triggered. The Interflow includes the directory to the file (`path`).

File Creation Anomaly

A file or files were created an anomalously large number of times. Check with the user to see if this is expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR File Anomaly (XT1003)
- Tags: [File Anomaly; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `file_creation`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- secondary

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `secondary` — user name
- `actual` — actual number of file creations in the period
- `typical` — typical number of file creations in the period
- `path` — path to the file(s) created

Use Case with Data Points

The number of file creations for each user (`command`) is calculated periodically. If the volume (`actual`) is much larger than the typical volume (`typical`) of file creations in any period, an alert is triggered. The Interflow includes the directory to the file (`path`).

Google Workspace Account Manipulation

A Google Workspace user was manipulated. Check with the user to make sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Account Anomaly (XT4007)
- Tags: [External; GSuite; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `gsuite_account_manipulation`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_detail.affected_email_address
- event_detail.name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_detail.affected_email_address` — key ID for the account
- `event_detail.name` — Google Workspace suspicious event name
- `event_detail.type` — Google Workspace suspicious event type

Use Case with Data Points

For each Google Workspace account (`event_detail.affected_email_address`), account manipulation is evaluated periodically. This alert is triggered if the Google Security center reports a leaked password or a user account being suspended for specific reasons. The Interflow includes the account ID (`event_detail.affected_email_address`), Google Workspace event name (`event_detail.name`), and Google Workspace event type (`event_detail.type`).

Google Workspace Attack Warning

Attacks to a Google Workspace account were observed. Check with the account holder.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [External; GSuite; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `gsuite_attack_warning`.

Severity

74

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_detail.name
- gsuite.actor.email
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `actor.email` — key ID for the account
- `srcip` — source IP address of the account
- `srcip_host` — source host name
- `event_detail.name` — Google Workspace suspicious event name
- `event_detail.type` — Google Workspace suspicious event type

Use Case with Data Points

For each Google Workspace account (`actor.email`), attacks are searched periodically. If an attack is identified, an alert is triggered. The Interflow includes the account ID (`actor.email`), source IP address (`srcip`), Google Workspace event name (`event_detail.name`), and Google Workspace event type (`event_detail.type`).

Google Workspace Suspicious Activities

Suspicious activities were observed in a Google Workspace account. Check with the account holder.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Login Anomaly (XT4006)
- Tags: [External; GSuite; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `gsuite_suspicious_activities`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_detail.name
- srcip
- user.name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `user.name` — key ID for the account
- `srcip` — source IP address of the account
- `srcip_host` — source host name
- `event_detail.name` — Google Workspace suspicious event name
- `event_detail.type` — Google Workspace suspicious event type

Use Case with Data Points

For each Google Workspace account (`actor.email`), suspicious activities are searched periodically. If suspicious activities are detected, an alert is triggered. The Interflow includes the account ID (`actor.email`), source IP address (`srcip`), Google Workspace event name (`event_detail.name`), and Google Workspace event type (`event_detail.type`).

Google Workspace User Suspended

A Google Workspace user was suspended. Check with the user to make sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Account Anomaly (XT4007)

- Tags: [External; GSuite; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `gsuite_user_suspended`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_detail.name
- gsuite.actor.email
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `actor.email` — key ID for the account
- `srcip` — source IP address of the account
- `srcip_host` — source host name
- `event_detail.name` — Google Workspace suspicious event name
- `event_detail.type` — Google Workspace suspicious event type

Use Case with Data Points

For each Google Workspace account (`actor.email`), suspension status is searched periodically. If a user is suspended, an alert is triggered. The Interflow includes the account ID (`actor.email`), source IP address (`srcip`), Google Workspace event name (`event_detail.name`), and Google Workspace event type (`event_detail.type`).

Hydra Password Guessing Hack Tool

A user from a Windows host executed a command-line script that launched either the `hydra.exe` command or a command using known Hydra parameters, which may be an inappropriate use of the Hydra password guessing tool.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Hydra; Brute Force; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `hydra_password_guessing_hack_tool`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip
- event_data.Image
- event_data.CommandLine

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — device internal IP address
- `computer_name` — name of the Windows host
- `event_data.CommandLine` — command used to run the tool
- `event_data.Image` — process running `hydra.exe` for password cracking

Use Case with Data Points

This alert is triggered if a Windows host (`hostip`) executes a PowerShell script with a context that includes one or more flags (`event_data.Image`or `event_data.CommandLine` indicating usage of the Hydra password guessing hack tool. The Interflow includes the IP address of the Windows host (`hostip`), the host name (`computer_name`), and the script image (`event_data.Image`) or script payload (`event_data.CommandLine`).

Validation / Remediation

Check the body of the Powershell script that is reported on the Windows host to identify whether the contents of the script are actually malicious. If malicious, consider quarantining the host.

Potential False Positives

The running of any executable named `hydra.exe` or a command that has parameters of `-u` and `-p` or `^user^` and `^pass^` triggers this alert.

ICMP Based Exfiltration or Tunneling 6.1.0

Possible ICMP based exfiltration activities have been observed.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Exfiltration (TA0010)
- Technique: Exfiltration Over Alternative Protocol (T1048)
- Tags: [ICMP; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `icmp_tunnel`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `dstip` — destination IP address
- `inbytes_delta` — total received bytes
- `inpkts_delta` — total received packets
- `outbytes_delta` — total sent bytes
- `outpkts_delta` — total sent packets
- `actual` — actual packet size transmitted through the tunnel in the time period
- `typical` — typical packet size transmitted through the tunnel in the time period

Use Case with Data Points

An anomalously large tunneled ICMP packet (`actual`) bytes has been observed between (`srcip`) and (`dstip`). The typical ICMP packet size transmitted through this tunnel is (`typical`) bytes, measured in a 5-minute interval. This indicates potential ICMP based exfiltration or tunneling.

Validation / Remediation

Check whether the abnormal ICMP packet is benign and whether the destination IP address is malicious.

Potential False Positives

New benign DNS server or CDN servers may utilize ICMP to transmit data.

For example, a user or IT script sending `ping -l 1400 8.8.8.8` (or `ping -s 1400` on Linux/MacOS) to test network MTU or connectivity may trigger this detection. These larger-than-normal ICMP packets can appear anomalous compared to the typical 32–64 byte payloads, especially if repeated within a short interval. However, this behavior is usually benign, particularly when the destination is a well-known diagnostic IP like Google DNS (`8.8.8.8`).


You can use an alert filter combined with a lookup list to filter false positives.

Impossible Travel Anomaly

A user logged in from locations that are geographically impossible to travel between in the time frame. Check with the user.


This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay

corresponding to their amount of latency.

The expected detection de is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

For the Impossible Travel Anomaly, there are two chances for ingestion delay, so the slowest of the two records will define the delay. This alert type is also sensitive to the order of user logins.

The Impossible Travel Anomaly employs frequent location alert suppression to address common false positive alert scenarios. If a user is known to frequently visit the travel destination, the alert will be suppressed.

You can customize the Impossible Travel Anomaly in the Detection Management System. See Configuring Customizations for ML Detections.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Location Anomaly (XT2001)
- Tags: [External; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `user_impossible_travel`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the source user
- `srcip_username` — source user name
- `srcip` — source IP address
- `srcip_host` — source host name
- `engid_gateway` — gateway IP address, used to determine the geo location when the source IP address is private
- `srcip_geo` — source IP address geo location, including latitude and longitude

- has_mfa — whether a Multi-Factor Authentication (MFA) event is associated with the current login 5.5.0
- distance_deviation — deviation in distance between two login locations (miles)
- time_deviation_human_readable— deviation in time between the two login events 6.2.0
- time_deviation — deviation in time between the two login events (seconds)
- travel_speed — calculated speed for the user to travel between two locations (miles/hour)
- appid_name — application name for the login event
- last_login_time — timestamp of event 2, the 2nd login
- _id2 — ID of event 2
- _index2 — index of event 2
- srcip2 — source IP address of event 2
- srcip_geo2 — source IP address geo location of event 2, including latitude and longitude
- has_mfa2 — whether an MFA event is associated with the previous login (event 2) 5.5.0

has_mfa and has_mfa2 are best-effort and are not guaranteed to be accurate, specifically when there are ingestion delays.

About srcip and srcip2: For any given travel, there is a source and a destination, which are defined by timestamp, so the source timestamp is less than the destination timestamp. Since the destination record arrives later than the source record, the destination is used as the basis for the alert. This means that srcip comes from the destination and srcip2 is added to that record from the source login. (This can be counterintuitive because one might assume that srcip2 comes after srcip.)

Use Case with Data Points

Login events (event 1 and event 2) are examined for a user (srcip_usersid), to see if the login locations (srcip_geo and srcip_geo2), that are at least 100 miles apart, changed faster (travel_speed = distance_deviation/time_deviation) than possible with the typical commercial flight speed of 600 miles/hour.

Event 1 is the basis for the Interflow. The srcip_usersid and srcip_username identify the user, appid_name identifies the application, and last_login_time identifies the time when the 2nd login event (event 2) happened. You can find detailed information about event 2 by checking id2 in index2, source IP (srcip2), and geo location (srcip_geo2).

Remove Duplicate Login Events for Microsoft 365 and Entra ID 6.2.0

When both Microsoft 365 and Entra ID data sources report the same login ID, Stellar Cyber will prioritize Entra ID for alert generation of login location anomalies, provided there is no significant ingestion delay. If Stellar Cyber can correlate the two logins by their login ID, the Entra ID record will be sent. The Entra ID records have metadata related to MFA logins as well as specific Azure

location enrichment, that the Microsoft 365 records do not have. The Entra ID record will still be prioritized even if it is not for MFA authentication.

However, if the Entra ID record is delayed, such as due to an ingestion delay, in order to maintain responsiveness, the Microsoft 365 record will be sent.

And if only Microsoft 365 is configured, those logs will continue to be used exclusively.

Using ASNs and User Agents to Improve Location Alert Triage 6.3.0

To improve triage of location-based alerts, Stellar Cyber will begin collecting user history for Autonomous System Number (ASN), ASN org, and user agents (if available from Microsoft 365), and then use this history to enrich alerts.

Enrichment fields are added to alerts detailing whether that user or tenant has seen that ASN, ASN org, or user agent before.

For the user and tenant scopes, the enriched fields are: `asn_seen_before`, `asn_org_seen_before`, `user_agent_seen_before`, and `location_seen_before`.

The user scope has an additional enriched field: `user_start_time_ms`, which is the time that tracking ASN and user agents started, in UNIX milliseconds.

> The `location_seen_before` field only tracks locations seen before the `user_start_time_ms`.

Using ASNs and user agents improves accuracy in anomaly detection and reduces false positives. You can also use the enriched fields for Automated Threat Hunting (ATH) or alert filtering.

Internal Account Login Failure Anomaly

An anomalously large number of login failures from an internal source IP address to an internal destination IP address was observed for an account. Check with the user.

This alert type has the following subtypes:

- Windows Logon Events
- Kerberos Events
- NTLM Events
- Hibun Security Logs

> This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay corresponding to their amount of latency.
>
> The expected detection delay is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)

- Technique: Brute Force (T1110)
- Tags: [Internal; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_cloud_account_login_failure`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — account user ID

  or

- `srcip_username` — account user name, enriched from `event_data.targetusername`

  The key field for this alert type can be either `srcip_usersid` or `srcip_username`, depending on the data feed.

- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `srcip_host` — host name of corresponding source IP address
- `login_type` — type of login
- `srcip_reputation` — source reputation

Use Case with Data Points

Login failures and successes between any internal IP addresses are calculated periodically for every account (`srcip_usersid`). If the number of failures is significantly larger than the number of successes, an alert is triggered. A sample Interflow includes the login type (`login_type`), source host (`srcip_host`), and source reputation (`srcip_reputation`).

Alert Subtype: Windows Logon Events 5.2.0

The Windows Logon Events alert subtype is the same as the Internal Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Windows logon events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_cloud_account_login_failure_win_logon`.

Alert Subtype: Kerberos Events 5.2.0

The Kerberos Events alert subtype is the same as the Internal Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Kerberos events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_cloud_account_login_failure_kerberos`.

Alert Subtype: NTLM Events 5.2.0

The NTLM Events alert subtype is the same as the Internal Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from NTLM events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_cloud_account_login_failure_ntlm`.

Alert Subtype: Hibun Security Logs 5.2.0

The Hibun Security Logs alert subtype is the same as the Internal Account Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Hibun security logs.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_cloud_account_login_failure_hibun`.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Brute-Forced Successful User Login

A successful login was observed from an IP address that had previously seen a large number of login failures, or a successful login to a user account was observed from another IP address or IP addresses that had previously seen a large number of login failures to that account. Check with the user.

This alert type has the following subtypes:

- Source IP Based
- User ID Based

This alert type has a relatively long detection delay of up to 40 minutes because it waits for login events from high latency data sources and is sensitive to the event order of user logins.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; Brute Force; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_user_success_brute_forcer`.

Severity

95

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- srcip_usersid

Alert Subtype: Source IP Based

The source IP-based alert subtype has the same **XDR Kill Chain** as the user ID-based alert subtype, but differs in the **Key Fields and Relevant Data Points** and **Use Case with Data Points**.

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_success_brute_forcer_srcip_usersid`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_usersid` — Windows SID associated with the source IP address

- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `source_geo.countryName` — source country
- `dstip_host` — destination host name
- `login_type` — type of login
- `username` — user name
- `related_alert._id` — link to the related Internal User Login Failure Anomaly

Use Case with Data Points

The login records to an internal IP address (`dstip`) are checked for every internal source IP address (`srcip`). An alert is triggered if that IP address:

1. Has so many failed login attempts that it triggered the **Internal User Login Failure Anomaly**, and
2. Had a successful login

A sample Interflow includes the source IP address (`srcip`), login type (`login_type`), source host name (`srcip_host`), source reputation (`srcip_reputation`), source country (`srcip_geo.countryName`), and user name (`username`).

Alert Subtype: User ID Based

The user ID-based alert subtype has the same **XDR Kill Chain** as the source IP-based alert subtype, but differs in the **Key Fields and Relevant Data Points** and **Use Case with Data Points**.

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_success_brute_forcer_srcip`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_usersid` — Windows SID associated with the source IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `source_geo.countryName` — source country
- `dstip_host` — destination host name
- `login_type` — type of login
- `username` — user name
- `related_alert._id` — link to the related Internal Account Login Failure Anomaly

Use Case with Data Points

The login records to a user account (`srcip_usersid`) are checked for every internal source IP address (`srcip`). An alert is triggered if that user account:

1. Has so many failed login attempts that it triggered the **Internal Account Login Failure Anomaly**, and
2. Had a successful login

A sample Interflow includes the source IP address (`srcip`), login type (`login_type`), source host name (`srcip_host`), source reputation (`srcip_reputation`), source country (`srcip_geo.countryName`), and user name (`username`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Credential Stuffing

An anomalously large amount of username/password testing was observed on an internal Windows authentication service. Check the activity after successful logins, and consider blocking the internal source IP addresses.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_credential_stuffing`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- msg_class

- service_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `msg_class` — `Microsoft-Windows-Security-Auditing` for Windows
- `service_id` — specific account ID of a service
- `login_failure_rate` — rate of login failures per minute in the period
- `unknown_users_rate` — rate of unknown user names per minute in the period
- `unknown_users_to_login_failures` — ratio of unknown user names to login failures in the period
- `suspicious_ips` — suspicious source IP addresses (up to 100)
- `possible_breached_ips` — list of malicious IP addresses that may have successful breach activities

Use Case with Data Points

Internal credential stuffing is the constant testing of username/password combinations on the AWS, Okta, or Windows authentication functions. Login activity is monitored and if the number of failed logins is larger than normal for that service, an alert is triggered. The Interflow includes the service (`msg_class`), tenant's account ID on that service (`service_id`), suspicious source IP address (`suspicious_ips`), login failure rate (`login_failure_rate`), unknown user rate (`unknown_users_rate`), the ratio of unknown users to login failures (`unknown_users_to_login_failures`), and a list of source IP addresses that might have suspicious activities and should be investigated (`possible_breached_ips`).

> *More on Internal versus External and Inbound vs Outbound...*
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Internal Exploited Vulnerability

A host with a vulnerability discovered by a security scanning tool was exploited by an attack on that same vulnerability, indicating a high probability of success. Check the target to see if it was compromised.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Exploited Vulnerability (XT2015)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_vuln_exploit_correlation`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- scan_record's srcip, dstip, ids.cve

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `tenantid` — tenant ID
- `vulnerability_id` — ID of the original security scan result
- `ids_event_id` — ID of the original IDS exploit event
- `srcip` (of security scan result) — IP address of the target `correlation_info.srcip`
- `dstip` (of IDS event) — IP address of the target (`correlation_info.dstip`)
- `srcip` (of IDS event) — IP address of the attacker (`correlation_info.srcip`)
- `vulnerability.cve` — CVE associated with the reported vulnerability
- `ids.cve` — CVE the attacker used to exploit the host

Use Case with Data Points

An attacker (`srcip`) with IP address A is performing an exploit against a target (`dstip`) with IP address B using a vulnerability (`ids.cve`) with CVE x. If any security scanning tool found the target (`srcip`) with IP address B to have a vulnerability (`vulnerability.cve`) with CVE x, an alert is triggered.

When an alert is triggered, a new correlation event is generated. The Interflow of the correlation event includes the ID of the IDS exploit event (`ids_event_id`), the ID of the security scan record (`vulnerability_id`), the IP address of the attacker (`correlation_info.srcip` of the IDS event), the IP address of the victim (`correlation_info.dstip` of the IDS event or `correlation_info.srcip` of the security scan record), and the CVE that was used in the exploit (`vulnerability.cve` and `ids.cve`).

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Internal Firewall Denial Anomaly

An internal source host had actions blocked by a firewall too many times. Investigate the firewall rules that were violated. If suspicious, block the internal source IP address.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Firewall Anomaly (XT2002)
- Tags: [Internal; Firewall Anomalies; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_fw_action`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_host

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address

- `srcip_host` — source host name
- `actual` — actual number of firewall denials in the period
- `typical` — typical number of firewall denials in the period
- `dstip_host` — host name of corresponding destination IP address
- `dev_name` — name of the firewall
- `engid_name` — name of the sensor

Use Case with Data Points

The number of firewall denials for every internal source IP address (`srcip`) is calculated periodically. If an internal source IP address's number of firewall denials (`actual`) is much larger than the historical count (`typical`) of all internal IP addresses, an alert is triggered. The Interflow includes the name of the firewall (`dev_name`), the name of the sensor (`engid_name`), and the destination host (`dstip_host`).

> *More on Internal versus External and Inbound vs Outbound...*
>
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Internal Firewall Policy Anomaly

In internal traffic, a firewall policy was triggered that has never been seen before (or has very rarely been seen). Investigate the policy, the intent, and the associated traffic to determine whether this activity is legitimate or malicious.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Firewall Anomaly (XT2002)
- Tags: [Internal; Firewall Anomalies; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_fw_policy_id`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- fw_policy_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `fw_policy_id` — ID of the violated firewall policy
- `days_silent` — number of days since this firewall policy was last seen
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `dev_name` — device name
- `dev_type` — device type
- `engid_name` — sensor name

Use Case with Data Points

A firewall policy ID (`fw_policy_id`), which is raised by a device (`dev_name` and `dev_type`) and captured by a sensor (`engid_name`), shows never seen or very rare (`days_silent`) traffic between an internal host (`srcip_host`) and another internal host (`dstip_host`). This will trigger an alert.

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
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> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Handshake Failure

There were too many handshake failures between two internal hosts, which might indicate port scanning. Check the source host to see if this was expected, and if not, consider blocking the host.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] Discovery (TA0007)
- Technique: Network Service Scanning (T1046)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_handshake_failure`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- appid_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the host with the handshake failures
- `srcip_host` — source host name
- `dstip` — destination IP address of the host with the handshake failures
- `dstip_host` — destination host name
- `timestamp` — when the scan happened

Use Case with Data Points

If an internal host (`srcip`) scans across many ports on another internal host (`dstip`), an alert is triggered. The Interflow includes the IP address of the potential attacker (`srcip`), the IP address of the victim (`dstip`), a special message flag (`msgtyp`), and when the scan happened (`timestamp`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
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have unique analytics for internal/external).

- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal IDS Signature Spike

A source IP address transmitted an anomalous number of different IDS signatures. Typically, this indicates host penetration or vulnerability scanning.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: Lateral Movement (TA0008)
- Technique: Exploitation of Remote Services (T1210)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_ids_signature_spike`.

Severity

65

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_summary.ids_signatures_summarize` — summarized IDS signatures
- `srcip_host` — source host name
- `actual` — actual number of unique IDS signatures in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of unique IDS signatures from the source IP address, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others

counted as 1

Use Case with Data Points

The number of unique IDS signatures (`ids.signature`) and severity (`ids.severity`), are calculated periodically. If many different exploits with unique IDS signatures are observed, an alert is triggered. The Interflow includes a source (`srcip`), timestamp, an accumulated severity of IDS signatures (`actual`), the usual accumulated severity of IDS signatures (`typical`), and a sampling of the IDS signatures used in the attack (`ids_signatures_summarize`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal IP / Port Scan Anomaly

A host has either generated an anomalous number of connections compared to the typical amount, or has triggered an anomalous number of connection failure responses, in the measured interval. This can indicate that an attacker is scanning for computers or ports to exploit. Check with the user.

This alert type has the following subtypes:

- Connection Failure Anomaly (Sensor Traffic)
- Connection Spike Anomaly (Firewall / Windows Traffic)

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] Discovery (TA0007)
- Technique: Network Service Scanning (T1046)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_port_scan`.

Severity



40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Alert Subtype: Connection Failure Anomaly (Sensor Traffic)

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `connection_failure_anomaly`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `num_failed` — unique number of (destination IP and destination port) tuples that respond with failed status
- `num_successful` — unique number of (destination IP and destination port) tuples that respond with success status
- `percent_failed` — percent of unique (destination IP and destination port) tuples that respond with failed status
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name
- `summary_port_scan_type` — port scan type 6.2.0

Example of Event Summarization in the Alert Description

Within a 5.0-minute period the overall traffic from 10.1.1.1 consisted of the following scans: 64.94% to 10.2.1.26/31 on port(s) ranging from 2 to 49157, 35.06% to 10.2.1.28/30 on port(s) ranging from 1 to 49157.

Use Case with Data Points

For each internal source IP address (`srcip`), the number of unique internal destination IP:port pairs that gave fail responses and the number of unique destination IP:port pairs that gave success responses are calculated periodically. If the number of failures is significantly larger than the number of successes, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination host (`dstip_host`), and application name (`appid_name`).

Validation / Remediation

Check with the user related to the internal source IP address. Inform the user's supervisor if the activity is unauthorized.

Potential False Positives

Some legitimate activities such  ulnerability scans or penetration testing may trigger this alert type.

Alert Subtype: Connection Spike Anomaly (Firewall / Windows Traffic)

Event Name

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `connection_spike_anomaly`.

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of connections to the destination IP address in the period
- `typical` — typical number of connections to the destination IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name
- `summary_port_scan_type` — port scan type 6.2.0

Example of Event Summarization in the Alert Description

Within a 5.0-minute period the overall traffic from 10.1.1.1 consisted of the following scans: 64.94% to 10.2.1.26/31 on port(s) ranging from 2 to 49157, 35.06% to 10.2.1.28/30 on port(s) ranging from 1 to 49157.

Use Case with Data Points

For every unique triplet (source IP address, destination IP address, and destination port) browsed by each source IP address (`srcip`), the number of response failures and successes and the number of total data volume are calculated periodically. If the number of failures is significantly larger than the number of successes, or the total data volume is significantly larger than the typical number, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination host (`dstip_host`), and application name (`appid_name`).

Considering that a lateral scan (private to private) is more sensitive than a non-lateral scan, this alert type is divided into two parts. One focuses on lateral scan analysis, the other focuses on non-lateral scan analysis. The mechanism remains the same as before, with the trigger condition for lateral scan alert being more sensitive than non-lateral one.

Validation / Remediation

Check with the user related to the internal source IP address. Inform the user's supervisor if the activity is unauthorized.

Potential False Positives

Some legitimate activities such as vulnerability scans or penetration testing may trigger this alert type.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Non-Standard Port Anomaly

An application had an anomalously large number of connections or a rarely seen connection to an internal IP address on non-standard ports. Check the application to be sure this is benign.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Service on Non-Standard Port (XT2011)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_non_std_port_anomaly`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- dstport
- appid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — destination IP address
- `dstport` — destination port
- `appid` — application ID
- `days_silent` — number of days since the application was last seen
- `appid_name` — application name
- `dstip_host` — host name of corresponding destination IP address
- `actual` — actual number of connections in the period
- `typical` — typical number of connections in the period

Use Case with Data Points

The number of connections for an application (`dst_ip` + `dstport` + `appid`) to an internal IP address is calculated periodically. If a non-standard combination has an actual number of connections (`actual`) that is much larger than the typical number of connections (`typical`), or the combination has not appeared for a long time, an alert is triggered. The Interflow includes the source host (`srcip_host`), destination IP address (`dstip`), destination port (`dstport`), application ID (`appid`), and application name (`appid_name`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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Internal Other Malware

Malware with uncategorized malicious activity in internal traffic was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR Malware (XTA0006)

- Technique: XDR Miscellaneous Malware (XT6001)
- Tags: [Internal; Malwa

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_malware_activity`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `ids.severity` — severity of the IDS signature 5.2.0
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the malware
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates malware in internal traffic that cannot be categorized as ransomware, spyware, trojan, PUA, or adware, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the malware (`file_name`) from the sandbox.

> *More on Internal versus External and Inbound vs Outbound...*
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as `inbound` or `outbound`. Use the following as a guide for these concepts:

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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Password Spraying

An anomalously large number of failed logins with unknown user names was observed on internal Windows authentication services. Check the activity after successful logins, and consider blocking the internal source IP addresses.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Sub-technique: Password Spraying (T1110.003)
- Tags: [Internal; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_password_spray`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_data.Workstation
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address generating a failed login

or

- `event_data.Workstation` — workstation generating a failed login (if applicable)

  

  The key field for this alert type can be either `srcip` or `event_data.Workstation`, depending on the data feed.

- `srcip_host` — source host name
- `event_id` — Windows event ID corresponding to the login failures
- `login_type` — type of login protocol; the available values vary by `event_id`
- `actual` — actual number of failed logins with unknown user names in a 5-minute period
- `typical` — typical number of failed logins with unknown user names in a 5-minute period
- `event_summary.password_spray_user_summary` — list of up to 100 unknown user names associated with the failed logins (the first three are shown in the alert description)

Use Case with Data Points

If a potential password spraying attack is observed, an alert is triggered. The Interflow includes a source (`srcip` or `event_data.Workstation`), timestamp, the type of login (`login_type`), the number of failed logins (`actual`), the usual number of failed logins (`typical`), and a sampling of the user names used in the attack (`password_spray_user_summary`).

More on Internal versus External and Inbound vs Outbound...

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- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal PII Leaked

Personally identifiable information (social security numbers or credit cards) has been observed in internal traffic in the clear. Check the source to see if it is compromised. If so, consider blocking it.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [Internal] Exfiltration (TA0010)
- Technique: Automated Exfiltration (T1020)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_pii_leak`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the PII leak
- `dstip` — destination IP address of the PII leak
- `ids.signature` — IDS signature of the exploit
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

If a personally identifiable information leak is detected by IDS, an alert is triggered. A sample Interflow includes the IDS signature (`ids.signature`), source IP address (`srcip`), destination IP address (`dstip`), source host (`srcip_host`), and destination host (`dstip_host`).

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>
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> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Plain Text Passwords Detected

A plain text password was observed in unencrypted traffic between internal systems. Check with the user.

This alert type looks for the presence of `metadata.request.password`, `metadata.request.auth_password`, and `metadata.request.auth_simple` in the Interflow records from the sensors. When plain text passwords are present in the network traffic, the sensors are able to decode and create the corresponding Interflow fields. To preserve privacy, the actual passwords are replaced by a sequence of asterisks (*).

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Clear Password (XT2006)
- Tags: [Internal; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_clear_password`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `actual` — actual number of connections with a plain text password in the period
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name

Use Case with Data Points

If there are plain text passwords in traffic records with a public source IP address (`srcip`) or destination IP address (`dstip`), an alert is triggered. A sample Interflow includes the source IP address (`srcip`), destination IP address (`dstip`), source host (`srcip_host`), destination host (`dstip_host`), and application (`appid_name`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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Internal Protocol Account Login Failure Anomaly

An anomalously large number of login failures between internal IP addresses over SMB or FTP was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_protocol_account_login_failure`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.username

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.request.username` — user name in the HTTP connection request
- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `appid_name` — application name
- `login_type` — type of login
- `srcip_host` — host name of corresponding source IP address
- `srcip_reputation` — source reputation

Use Case with Data Points

For every user name (`metadata.request.username`) in the HTTP connections names (that do not begin with "Mozilla" or "Aella"), the number of failed and successful logins are calculated periodically. If the number of failed logins is much greater than successful logins, an alert is triggered. The Interflow includes the application name (`appid_name`), login type (`login_type`), source host (`srcip_host`), and source reputation (`srcip_reputation`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

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- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Protocol Pattern Anomaly 6.2.0

An uncommon pattern of applications usage was observed. Investigate if the source host has unexpected application connections. It could be a malware infected host.

This alert type focuses on lateral movement protocols (ssh, smb, rdp, ldap, ldaps, msrpc, dcerpc, telnet, teamviewer, krb5, pcanywhere, anydesk, realvnc, and wmi).

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Remote Services (T1021)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_appflow_outlier`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- appid_name
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `appid_name` — name of the internal protocol
- `dstip_type` — IP address type of the connection destination
- `num_clust` — number of common protocol patterns observed
- `sum_clust_size` — number of source IP addresses that have a similar protocol pattern
- `training_days` — number of days of protocol pattern data

Use Case with Data Points

The usage pattern from (`srcip`) of (`dstip_type`) (`appid_name`) is unusual compared to other (`num_clust`) different types of usage patterns observed from (`sum_clust_size`) source IP addresses in (`training_days`) days.

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).

- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal PUA

Unwanted applications or malware that bombards the user with advertisements in internal traffic has been observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR Malware (XTA0006)
- Technique: XDR PUA (XT6002)
- Tags: [Internal; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_pua`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)

- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the PUA
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates potentially unwanted applications (PUA) in internal traffic, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the PUA (`file_name`) from the sandbox.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Ransomware

Malware that prevents you from accessing your system or files and demands ransom payment in order to regain access in internal traffic was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [Internal] Impact (TA0040)
- Technique: Data Encrypted for Impact (T1486)
- Tags: [Internal; Malware; Ransomware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_ransomware`.

Severity

98

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the ransomware
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates ransomware in internal traffic, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the ransomware (`file_name`) from the sandbox.

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.

- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal RDP BlueKeep

The use of a scanner by zerosum0x0 that discovers targets vulnerable to BlueKeep (CVE-2019-0708) has been observed between internal hosts. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Privilege Escalation (TA0004)
- Technique: Exploitation for Privilege Escalation (T1068)
- Tags: [Internal; RDP; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_rdp_bluekeep`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- ids.signature
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `srcip_host` — source host name
- `dstip_host` — destination host name

Use Case with Data Points

If the scanner by zerosum0x0 is used, an alert is triggered. A sample Interflow includes the IDS signature (`ids.signature`), source host (`srcip_host`), and destination host (`dstip_host`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal RDP Brute Force Attack

An anomalously large number of RDP connections from internal host(s) to an RDP server were observed. Check the source IP addresses to see if they are unknown or malicious, and monitor any successful RDP logins.

XDR Kill Chain

- Kill Chain Stage:Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; RDP; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_rdp_brute_force`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — IP address of the destination RDP server
- `dstip_host` — destination host name

- `actual` — actual number of RDP connections to the destination IP address in the period
- `typical` — typical number of RDP connections to the destination IP address in most time buckets
- `srcip` — source IP address
- `srcip_host` — source host name

Use Case with Data Points

RDP connection activity is monitored and the number of connections calculated periodically. If the number of connections from internal host(s) to an RDP server (`actual`) is much greater than normal (`typical`), an alert is triggered. A sample Interflow includes the destination IP address (`dstip`) and source IP address (`srcip`).

> *More on Internal versus External and Inbound vs Outbound...*
>
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> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal RDP Suspicious Outbound

Non-standard tools from an internal host connecting to TCP port 3389 in the other internal host were observed. This could indicate lateral movement attempting to establish an RDP connection. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Remote Services (T1021)
- Tags: [Internal; RDP; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_rdp_suspicious_outbound`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the host that connects to TCP port 3389 with a non-standard tool
- `srcip_host` — source host name
- `process_name` — process name

Use Case with Data Points

Connections to TCP port 3389 are monitored, and if non-standard tools connect, an alert is triggered. A sample Interflow includes the source IP address (`srcip`) and the process name (`process_name`). The following are the standard tools:

- mstsc.exe
- RTSApp.exe
- RTS2App.exe
- RDCMan.exe
- ws_TunnelService.exe
- RSSensor.exe
- RemoteDesktopManagerFree.exe
- RemoteDesktopManager.exe
- RemoteDesktopManager64.exe
- mRemoteNG.exe
- mRemote.exe
- Terminals.exe
- spiceworks-finder.exe
- FSDiscovery.exe
- FSAssessment.exe
- MobaRTE.exe
- chrome.exe
- thor.exe
- thor64.exe

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Scanner Behavior Anomaly

An anomalously large amount of scanning behavior or a rarely seen scan behavior between internal hosts was observed. Cross-check with the **IP / Port Scan Anomaly** alert.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] Discovery (TA0007)
- Technique: Network Service Scanning (T1046)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_scan_anomalies`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — signature of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as

1

- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name

Use Case with Data Points

The number of occurrences of each scanner, based on IDS signature (`ids.signature`) between internal hosts, is calculated periodically. If one scanner occurs (`actual`) much more often compared to its history (`typical`), an alert is triggered. A sample Interflow is presented with information such as the traffic application type (`appid_name`), source host (`srcip_host`), and destination host (`dstip_host`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal SMB Read Anomaly

An internal IP address sent an anomalously large number of read requests to an internal SMB protocol based service(s). Investigate the files that this internal IP address tried to read. If suspicious, block the specific internal source IP address.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Exploitation of Remote Services (T1210)
- Tags: [Internal; SMB; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_smb_read_anomaly`.

Severity

20 

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of SMB reads from the source IP address in the period
- `typical` — typical number of SMB reads from other source IP addresses in the period
- `dstip_host` — destination host name
- `smb_username` — SMB user name
- `event_summary.smb_path_list` — folders experiencing a high volume of SMB read requests (the first three are shown in the alert description) 5.2.0

Use Case with Data Points

The number of SMB read requests for every internal source IP address (`srcip`) is calculated periodically. If a source IP address's number of SMB reads (`actual`) is much larger than the typical number (`typical`) and that of other IP addresses in any period, an alert is triggered. The Interflow includes the SMB user (`smb_username`) and destination host (`dstip_host`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal SMB Username Enumeration

At least 5 different users SMB l attempts and 1 denied attempt or at least 10 different users SMB login attempts, were observed from an internal IP address to other internal IP address(es). Check the source IP address. If malicious, consider blocking it.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; SMB; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_smb_user_scan`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `actual` — actual unique SMB user count
- `typical` — SMB user count threshold
- `smb_username_set` — all SMB login user names

Use Case with Data Points

If an internal source IP address (`srcip`) has several SMB login attempts with (1) at least 5 unique user names and at least 1 denied attempt or (2) at least 10 unique user names, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), destination host (`dstip_host`), and all the user names (`smb_username_set`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).

- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal SMB Write Anomaly

An internal IP address sent an anomalously large number of SMB write requests to other internal IP address(es). Investigate the files that the IP address tried to write. If suspicious, block the source IP address.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Remote Services (T1021)
- Tags: [Internal; SMB; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_smb_anomaly`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — source host name
- `actual` — actual number of SMB writes in the period
- `typical` — typical number of SMB writes in the period
- `dstip_host` — destination host name
- `smb_username` — SMB user name
- `event_summary.smb_path_list` — folders experiencing a high volume of SMB write requests (the first three are shown in the alert description) 5.2.0

Use Case with Data Points

The number of SMB write requests to internal IP address(es) for every internal source IP address (`srcip_host`) is calculated periodically. If a source IP address's number of SMB writes (`actual`) is much larger than the typical number (`typical`) and that of other IP addresses in any period, an alert is triggered. The Interflow includes the SMB user (`smb_username`) and destination host (`dstip_host`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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Internal Spyware

Malware that collects and shares information about a device without consent in internal traffic was observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR Malware (XTA0006)
- Technique: XDR Spyware (XT6003)
- Tags: [Internal; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_spyware_activity`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature
- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the spyware
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates spyware activity in internal traffic, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the spyware (`file_name`) from the sandbox.

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
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> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction

of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal SQL Anomaly

An internal IP address sent an anomalously large number of queries to an internal SQL server. Investigate the queries. If suspicious, block the source IP address.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Exploitation of Remote Services (T1210)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_mysql_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `actual` — actual number of SQL queries in the period
- `typical` — typical number of SQL queries from the source IP address
- `dstip_host` — destination host name

Use Case with Data Points

The number of SQL queries for every internal source IP address (`srcip_host`) is calculated periodically. If an internal source IP's SQL query count (`actual`) is much larger than the typical count (`typical`) and that of other internal IP addresses in any period, an alert is triggered. The internal source IP's country is (`srcip_geo.countryName`). The Interflow includes the internal destination host (`dstip_host`) the source IP visits.

> *More on Internal versus External and Inbound vs Outbound...*
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- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
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- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal SQL Dumpfile Execution

The SQL `dumpfile` command between two internal IP addresses was observed. This command is commonly used to dump database content or query output to a file on disk. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] Collection (TA0009)
- Technique: Data Staged (T1074)
- Tags: [Internal; Database; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_sql_db_dump`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `actual` — number of SQL `dumpfile` queries
- `srcip_host` — source host name
- `source_geo.countryName` — source country

- `dstip_host` — destination host name

Use Case with Data Points

If any SQL `dumpfile` commands are detected between an internal source IP address (`srcip`) and an internal destination IP address (`dstip`), an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), and the number of SQL `dumpfile` queries in the period (`actual`).

More on Internal versus External and Inbound vs Outbound...

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- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal SQL Shell Command

Shell commands were observed over a SQL connection, which is a common way hackers try to gain shell access over vulnerable SQL applications. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [Internal] Execution (TA0002)
- Technique: Command and Scripting Interpreter (T1059)
- Tags: [Internal; Database; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_database_command`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `dstip` — destination IP address
- `srcip_host` — source host name
- `srcip_reputation` — source reputation
- `dstip_host` — destination host name
- `dstip_reputation` — destination reputation
- `metadata.request.query` — SQL query command
- `actual` — number of query records from one source to one destination in one period

Use Case with Data Points

For SQL query records, if special commands (such as `select mylab_sys_exec`) are found, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), destination IP address (`dstip`), source host (`srcip_host`), source reputation (`srcip_reputation`), destination host (`dstip_host`), destination reputation (`dstip_reputation`), and SQL query records (`metadata.request.query`).

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Internal Suspected Malicious User Agent

An internal HTTP connection was made by a user agent that has been identified as potentially malicious. Investigate the connection's destination.

This alert type has the following subtypes:

- Predicted Malicious Agent
- Known Malicious Agent Match

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR User Agent Anomaly (XT2012)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_suspected_malicious_user_agent`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.user_agent

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.request.user_agent` — user agent in the HTTP connection request
- `user_agent.original` — user agent in the HTTP connection request
- `stellar.confidence` — model's confidence in the prediction used to make the alert
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name

Use Case with Data Points

If a seen user agent is identified as suspicious, an alert is triggered. The alert will contain the suspicious user agent (`metadata.request.user_agent`), confidence (`stellar.confidence`), tenant (`tenant_name`), source IP (`srcip`), and destination IP (`dstip`) in the key fields. Additionally, the confidence level of the model is displayed in the alert description in a pop-up box.

Alert Subtype: Predicted Malicious Agent

The Predicted Malicious Agent alert subtype is the same as the Internal Suspected Malicious User Agent alert type above, with the following differences:

- The `stellar.anomaly_tag` is `predicted_internal`.

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_suspected_malicious_user_agent`.
- It is triggered by a machine learning classifier.

Alert Subtype: Known Malicious Agent Match

The Known Malicious Agent Match alert subtype is the same as the Internal Suspected Malicious User Agent alert type above, with the following differences:

- The `stellar.anomaly_tag` is `known_internal`.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_suspected_malicious_user_agent_known_malicious`.
- It is triggered by known threats.

More on Internal versus External and Inbound vs Outbound...

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- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal Suspected SQL Injection 6.4.0 6.6.0

A source host is exhibiting internal SQL injection behavior with multiple queries observed in the last 60 minutes.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Exploitation of Remote Services (T1210)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_http_sqli`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — source host name
- `dstip_host` — destination host name
- `dstport` — destination port
- `event_summary.sqli.top_sqli_uris` — suspected queries from source host observed in time window (top 5 shown when more than 5 are detected)
- `stellar.sqli_variation` — unique queries from the source host observed in the time window
- `stellar.sqli_duration` — time window for suspicious queries observation from source host

Use Case with Data Points

An internal host is attacking your web services with SQL injection attacks using many unique payloads observed in the last 60 minutes. Consider disconnecting this host and check for malicious programs.

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Internal SYN Flood Attacker

An internal attacker sends a large amount of SYN requests to internal target system(s) in an attempt to consume enough server resources to make the system unresponsive to legitimate traffic.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [Internal] Impact (TA0040)
- Technique: Endpoint Denial of Service (T1499)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_syn_flood_attacker`.

Severity

25

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the SYN flood
- `srcip_host` — source host name
- `dstip` — target IP address of the SYN flood
- `dstip_host` — destination host name
- `dstport` — port on the target host that received the SYN flood
- `syn_flood_events` — number of SYN packets during the period

Use Case with Data Points

If an internal host (`srcip`) sends too many SYN packets (`syn_flood_events`) to internal target(s) (`dstip`) in a five-minute time window, an alert is triggered. The Interflow includes the IP address of the source host (`srcip`), the IP address of the target host (`dstip`), the port of the target host (`dstport`), and how many SYN packets were observed (`actual`).

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>
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Internal SYN Flood Victim

A large amount of SYN requests to an internal target were observed, which can indicate an attempt to consume server resources and make the target unresponsive. Check to see if the host is malicious or compromised. If so, consider blocking the source host.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [Internal] Impact (TA0040)
- Technique: Endpoint Denial of Service (T1499)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_syn_flood`.

Severity

25

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address for the SYN flood
- `srcip_host` — source host name
- `dstip` — target IP address of the SYN flood
- `dstip_host` — destination host name
- `dstport` — port on the target host that received the SYN flood
- `syn_flood_events` — number of SYN packets during the period

Use Case with Data Points

If an internal host (`srcip`) sends too many SYN packets (`syn_flood_events`) to internal target(s) (`dstip`) in a five-minute time window, an alert is triggered. The Interflow includes the IP

address of the source host (`srcip`), the IP address of the target host (`dstip`), the port of the target host (`dstport`), and how many SYN packets were observed (`actual`).

More on Internal versus External and Inbound vs Outbound...

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Internal Trojan

Malware that disguises itself as legitimate software in order to gain access to a system or files in internal traffic has been observed. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR Malware (XTA0006)
- Technique: XDR Trojan (XT6004)
- Tags: [Internal; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_trojan_activity`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip
- fileid
- ids.signature

- maltrace-cloud.data.malicious_activity
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — IDS signature
- `maltrace-cloud.data.malicious_activity` — malicious activity
- `actual` — number of records for one IDS signature or malicious activity in the period
- `lateral` — boolean, indicating whether this activity is lateral (from private to private)
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country
- `file_name` — name of the file that carries the trojan
- `event_source` — source of the event, either `ids` or `sandbox`

Use Case with Data Points

If ML-IDS or sandbox indicates trojan activity in internal traffic, an alert is triggered. A sample Interflow includes malicious activity for sandbox (`maltrace-cloud.data.malicious_activity`), IDS signature for ML-IDS (`ids.signature`), event source (`event_source`), source host (`srcip_host`), source country (`srcip_geo.countryName`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and the name of the file that carries the trojan (`file_name`) from the sandbox.

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>
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Internal URL Reconnaissance Anomaly

An anomalous number of HTTP 4xx errors from an internal IP address to other internal IP addresses were observed. This can indicate an attacker scanning for pages to exploit. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] Discovery (TA0007)
- Technique: Network Service Scanning (T1046)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_url_scan`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `event_summary.total_failed` — number of unique URLs with HTTP error status response in the period
- `event_summary.total_successful` — number of unique URLs with HTTP success status response in the period
- `event_summary.total_fail_ratio` — percent of unique URLs with HTTP error status response in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `srcip_geo.countryName` — source country name

Use Case with Data Points

For each internal source IP address (`srcip`), the number of unique URLs that responded with failure HTTP status and the number of unique URLs that responded with success HTTP status are calculated periodically. If the fail metric is significantly larger than the success metric, an alert is triggered. A sample Interflow includes the source host (`srcip_host`), destination host (`dstip_host`), and source country (`srcip_geo.countryName`).

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Internal User Application Usage Anomaly

An internal user who usually runs a few applications with internal service IP addresses suddenly runs a new application. Investigate the application to see if it is benign. Check with the user to see if this was expected.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR UBA (XTA0004)
- Technique: XDR App Anomaly (XT2003)
- Tags: [Internal; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_user_uncommon_app`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — source user ID
- `appid_name` — application name

- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_family` — application family
- `srcip_username` — source user name
- `stability` — score measuring the time since the last new application was used
- `days_stable` — time since the last new application was used
- `diversity` — score measuring the number of applications that the user used
- `child_count` — number of applications that the user used

Use Case with Data Points

An alert is triggered under the following conditions:

- a user (`srcip_usersid`, `srcip_username`) with a small number of applications (`diversity`, `child_count`) who has not used a new application for a long period of time (`stability`, `days_stable`), and then
- a new application (`appid_name`) belonging to an application family (`appid_family`) appears on a host (`scrip_host`) with this user, and
- that host connects to another host (`scrip_host`)

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal User Data Volume Anomaly

A user had an anomalously large volume of internal traffic compared to its typical volume or that of its peers. Investigate the user to determine if this is expected.



Firewall and non-firewall data do not contribute to the same alert, so this alert will have either entirely firewall data or no firewall data.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR (XTA0004)
- Technique: XDR Bytes Anomaly (XT3001)
- Tags: [Internal; User Behavior Analytics; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_user_bytes_sum`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — source user ID
- `actual` — actual traffic volume in the period
- `typical` — typical traffic volume from the user
- `srcip_host` — host name of corresponding source IP address
- `srcip_username` — source user name
- `dstip_host` — host name of corresponding destination IP address
- `dstip_reputation` — destination reputation
- `dstip_geo.countryName` — destination country
- `appid_name` — application name

Use Case with Data Points

The total internal traffic volume of each user identified by user ID (`scrip_usersid`) is calculated periodically. If the volume in one period (`actual`) is much larger than its normal volume (`typical`), an alert is triggered.

The Interflow includes the source IP address (`srcip_host`), destination IP address (`dstip_host`), destination reputation (`dstip_reputation`), destination country (`dstip_geo.countryName`), and application of the traffic (`appid_name`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.

- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Internal User Login Failure Anomaly

An anomalous number of login failures between internal IP addresses was observed for one of the following applications: SSH, SMTP, FTP, RDP, SMB, databases, Active Directory, Office 365, Okta, AWS CloudTrail, Google Workspace, Salesforce, or Microsoft Entra ID (formerly Azure Active Directory). Check with the user.

This alert type has the following subtypes:

- Source IP Based
- Destination IP Based
- NTLM Events
- Kerberos Events
- Windows Logon Events

This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay corresponding to their amount of latency.

The expected detection delay is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `internal_user_login_fail`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `service_id` — source domain, workstation, organization, or service
- `dstip` — destination IP address
- `dstip_host` — destination host name
- `event_summary.total_failed` — number of failed logins in the period
- `event_summary.total_successful` — number of successful logins in the period
- `event_summary.total_fail_ratio` — percent of failed logins in the period, which is: `event_summary.total_failed` / (`event_summary.total_failed` + `event_summary.total_successful`)
- `weighted_anomaly_score` — net score based on weighted rating of successful versus failed attempts (scanning, login, or other). Scores greater than the detection threshold will generate an alert.
- `login_type` — type of login, such as `ssh_traffic`, `okta_log`, or `aws_cloudtrail`
- `srcip_host` — source host name
- `srcip_reputation` — source reputation

Use Case with Data Points

Login failures and successes between internal IP addresses are calculated periodically for every source (`srcip`) and destination (`dstip`) IP address. If the number of failures is significantly larger than the number of successes, an alert is triggered. The Interflow includes the login type (`login_type`), source host (`srcip_host`), and source reputation (`srcip_reputation`).

Alert Subtype: Source IP Based 5.2.0

The Source IP-based alert subtype is the same as the Internal User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from network traffic, system logs, Linux events, and AWS events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_login_fail_srcip`.

Alert Subtype: Destination IP Based 5.2.0

The Destination IP-based alert subtype is the same as the Internal User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from network traffic, system logs, Linux events, and AWS events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_login_fail_dstip`.

Alert Subtype: NTLM Events 5.2.0

The NTLM Events alert subtype is the same as the Internal User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from NTLM events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_login_fail_ntlm`.

Alert Subtype: Kerberos Events 5.2.0

The Kerberos Events alert subtype is the same as the Internal User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Kerberos events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_login_fail_kerberos`.

Alert Subtype: Windows Logon Events 5.2.0

The Windows Logon Events alert subtype is the same as the Internal User Login Failure Anomaly alert type above, with the following differences:

- The subtype is for data sources from Windows Logon events.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `internal_user_login_fail_win_logon`.

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Login Attempt Location Count 6.1.0

A user has login attempts from an unusually high number of locations within a 24-hour period.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR Anomaly (XTA0004)
- Technique: XDR Location Anomaly (XT2001)
- Tags: [External; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `login_location_count`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — source user ID where logins were attempted
- `actual` — number of distinct login attempt locations. Near-duplicates, locations that are similar, are merged using geohashing.
- `typical` — Typical number of login attempt locations
- `event_summary.login_location_count.location_summary` — enumerated locations. This is an event summary for logins from this user over the last 24 hours. Near-duplicates may be considered identical by the alert type but are separated here. If the user has a very high number of logins, not all of them will be shown here.

> The number of locations displayed in `actual` may be less than the number of locations displayed in `location_summary`. These fields process the number of locations differently, which may result in a lower number for `actual` than the fully enumerated locations in `location_summary`.
>
> In Stellar Cyber, geohashing is a method of converting latitude/longitude coordinates into a short alphanumeric code that represents a geographic area. With a precision of 4, each geohash corresponds to a grid cell roughly 25 miles in diameter. Locations within the same cell share the same geohash, allowing the system to group near-duplicate points and reduce noise caused by unstable or inconsistent location enrichment. If two locations fall near a geohash boundary, they may be assigned different codes and treated as separate. This grouping is applied only for display and summarization purposes; the full list of unique enriched locations is still available in the alert.

Use Case with Data Points

A user (`srcip_usersid`) has experienced login attempts from an unusually high number of locations (`actual`) over the last 24 hours. This covers certain types of brute-forced login attempts and login location patterns that are not captured by other detections.

Validation / Remediation

Investigate the sources of login that user to identify if this behavior is expected.

Potential False Positives

N/A

Login Time Anomaly

A user logged in at an abnormal time. Check with the user.

> This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay corresponding to their amount of latency.
>
> The expected detection delay is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

This alert type reads the System Timezone in **Global Settings** and puts the timezone into the alert descriptions. (In **Global Settings**, set your timezone relative to UTC.)

When a Login Time Anomaly occurs, the timezone is bound to the alert description with the following priorities:

- The timezone inferred from `engid_gateway` takes precedence over the DP timezone, but only when it is present. If `engid_gateway` is present, the description will use the timezone where the login actually happened.
- If `engid_gateway` is not present, the DP timezone setting is used.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Time Anomaly (XT4005)
- Tags: [External; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `user_login_time`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID of the source user

  or

- `event_data.TargetUserName` — name of the user (Windows event)

-  The key field for this alert type can be either `srcip_usersid` or `event_data.TargetUserName`, depending on the data feed.

- `srcip_username` — source user name
- `srcip_host` — host name of corresponding source IP address
- `srcip_geo.countryName` — source country
- `actual_range` — actual login time range
- `typical_range` — typical login time range

Use Case with Data Points

Every user's (`srcip_usersid`) login time (`actual`) is compared to the typical login times (`typical_range`). If it is outside the range, an alert is triggered. The Interflow includes information such as the source user name (`srcip_username`), source host name (`srcip_host`), and source country (`srcip_geo.countryName`), as well as the destination host (`dstip_host`).

Long App Session Anomaly

An application had an anomalously long session compared to its typical session length or that of its peers. Investigate the application to see if this session was expected.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Session Anomaly (XT2005)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `long_session_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- appid_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `appid_name` — application name
- `actual` — actual maximum session length in the period
- `typical` — typical session length from the application's own history
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

Every application's (`appid_name`) maximum session duration is calculated periodically. If an application's maximum duration (`actual`) is much larger than its normal value (`typical`) or the typical value of other applications, an alert is triggered. The Interflow includes the source host (`srcip_host`) and destination host (`dstip_host`).

Malicious Site Access

A host accessed a URL with a reputation for potentially hosting malware. Check the URL and, if malicious, consider blocking it. Check the host for compromise.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Bad Reputation (XT2010)
- Tags: [External; Network Traffic Analysis; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `mal_access`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- url

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the host that initiated the site access
- `srcip_host` — source host name
- `url` — URL that was accessed
- `url_reputation` — reputation of the accessed URL

Use Case with Data Points

When a host (`srcip`) accesses a URL with a reputation (`srcip_reputation`) as potential malware hosting (`MalAccess`), an alert is triggered. The Interflow includes the source host IP address (`srcip`), the URL accessed (`url`), and the reputation of the URL (`url_reputation`).

Malware on Disk

Malicious software or a potentially unwanted application was found on a device and reported as not cleaned. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR Malware (XTA0006)
- Technique: XDR Miscellaneous Malware (XT6001)
- Tags: [Internal; Malware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `malware_on_disk`.

Severity

90 (Windows Defender)

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip
- file_path

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host
- `file_path` — file path
- `computer_name` — computer name
- `malware_engine` — malware engine, `Windows Defender`
- `group` — type of malware
- `type` — status of malware

Use Case with Data Points

If either of the following occurs, an alert is triggered:

- Windows Defender indicates a failure or error when taking actions to protect the system

A sample Interflow includes the computer name (`computer_name`), malware engine (`malware_engine`), host IP address (`hostip`), and path to the file (`file_path`).

Microsoft Entra Apps Modified to Allow Multi-Tenant Access

Microsoft Entra ID (formerly Azure Active Directory) observed an application being modified to allow multi-tenant access. Check with the organization to be sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold

- Tactic: Persistence (TA0003)
- Technique: Account Manipulation (T1098)
- Tags: [Microsoft Entra; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `azure_ad_add_app_multitenant`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — user account that modified the property change
- `activityDisplayName` — description of the action
- `targetResources.modifiedProperties.displayName` — properties that were changed (Display Name, New Value, Old Value)

Use Case with Data Points

If Microsoft Entra ID detects any user (`srcip_usersid`) changing an application to allow multi-tenant access, an alert is triggered. A sample Interflow includes the user ID (`srcip_usersid`), activity name (`activityDisplayName`), and name of the changed property (`targetResources.modifiedProperties.displayName`).

Microsoft Entra Custom Domains Changed

Microsoft Entra ID (formerly Azure Active Directory) observed a custom domain being changed. Check with the organization to be sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Domain Policy Modification (T1484)
- Tags: [Microsoft Entra; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `azure_ad_change_domain`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — user account that made the domain change
- `activityDisplayName` — activity display name
- `activity_name` — action description
- `targetResources.modifiedProperties` — properties that were changed (Display Name, New Value, Old Value)

Use Case with Data Points

If Microsoft Entra ID detects any user (`srcip_usersid`) changing a custom domain, an alert is triggered. A sample Interflow includes the user ID (`srcip_usersid`) and activity name (`activity_name`).

Mimikatz Credential Dump

A potential Mimikatz memory dump was observed. Check the process to determine whether the host is compromised. Consider quarantining the host.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: OS Credential Dumping (T1003)
- Tags: [Internal; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `mimikatz_mem_scan`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- access_subject
- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name

- `access_subject` — process attempting access
- `access_mask` — mask of the suspicious process used to obtain access privileges (different access masks indicate different capabilities obtained by the suspicious process)

Use Case with Data Points

If a process (`access_subject`) on a Windows host (`srcip`) tries to access `lsass.exe` with a special access mask (`access_mask`), an alert is triggered. The Interflow includes the IP address of the Windows host (`srcip`), the process performing mimikatz activity (`access_subject`), and the access mask used to acquire access privilege (`access_mask`).

Rule Based Detection

The Mimikatz Credential Dump rules are used to identify suspicious activity relating to potential Mimikatz memory dump. Any one or more of these will trigger the Mimikatz Credential Dump alert type.

The rule-based detection is the same as the Mimikatz Credential Dump alert type above, with the following difference:

- The `detection_flag` is a Stellar Cyber enriched field:
  - **2301:** Mimikatz access to lsass.exe

Link to Rule-Based Alert Types

Rules Contributing to Mimikatz Credential Dump Alert Type

Mimikatz DCSync

An attempt to replicate Active Directory for the first time on a domain controller, or the first time by that account, has occurred. Evaluate whether the replication is intended and, if not, consider disabling the account involved in the replication and investigate for further signs of compromise.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: OS Credential Dumping (T1003)
- Tags: [Internal, Active Directory; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `mimikatz_dcsync`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name
- event_data.SubjectUserSid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the targeted domain controller
- `hostip_host` — host name of the targeted domain controller
- `event_data.SubjectDomainName` — domain of the account that attempted the Active Directory replication
- `event_data.SubjectUserName` — name of the account that attempted the Active Directory replication

Use Case with Data Points

This alert is triggered when replication of an Active Directory domain controller (`hostip`) occurs for the first time or is attempted by a user account or computer account (`event_data.SubjectUserName`) that has rarely occurred (`days_silent`) or never initiated replication on that DC before. The Interflow includes the IP address of the targeted domain controller (`hostip`), the account (`event_data.SubjectUserName`) attempting the replication and its domain (`event_data.SubjectDomainName`), and the replication operation attempted (`event_data.Properties`). (For guidance understanding the GUID in the event_data.Properties field, refer to Microsoft Documentation.)

Validation / Remediation

To triage an alert of this type, consider first verifying whether the Active Directory replication event was expected. If the replication is not intended, then the alert has indicated that a DCSync attack is highly likely. This attack can be very severe, because all password hashes stored on the targeted domain controller might have been dumped. Disable the account involved in the replication as soon as possible and further investigate the account for any signs of compromise.

There is no simple remediation for a confirmed DCSync attack. Evaluate the overall risks of credential leakage and apply appropriate corrective actions, including minimizing accounts with permissions to perform Active Directory replication, and forcing a change of credentials for accounts with weak passwords.

Potential False Positives

The following will trigger an alert:

- Set up of a new DC
- Replication of a DC for the first time

Office 365 Admin Audit Logging Disabled

Office 365 admin audit logging was disabled. Make sure this change was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Impair Defenses (T1562)
- Sub-technique: Disable Cloud Logs (T1562 .008)

- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_admin_audit_logging_disabled`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationName` — organization with audit logging

Use Case with Data Points

Office 365 monitors each Office 365 account (`srcip_usersid`) for admin audit logging status. If admin audit logging is disabled, an alert is triggered. A sample Interflow includes the account ID (`srcip_usersid`) and organization name (`OrganizationName`).

Office 365 Content Filter Policy Changed

The Microsoft Exchange content policy was changed. An overly permissive content policy can allow spammers to send your organization unwanted email. Make sure this change was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Persistence (TA0003)
- Technique: Account Manipulation (T1098)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_content_filter_policy_changed`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationId` — ID of the organization with the Microsoft content policy change
- `OrganizationName` — organization with the Microsoft content policy change

Use Case with Data Points

Office 365 monitors all Office 365 accounts (`srcip_usersid`) in each organization (`OrganizationId`) for a Microsoft Exchange content policy change. If a change is detected, an alert is triggered. A sample Interflow includes the Office 365 account ID (`srcip_usersid`), organization ID (`OrganizationId`), and organization name (`OrganizationName`).

Office 365 File Sharing with Outside Entities

An Office 365 account shared multiple files with entities outside of the organization. Check with the user to make sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Exfiltration (TA0010)
- Technique: Transfer Data to Cloud Account (T1537)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_outside_entity_file_sharing`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `srcip` — source IP address of the sharing action
- `srcip_host` — source host name
- `srcip_geo.countryName` — source country

Use Case with Data Points

Office 365 monitors sharing with outside entities for each Office 365 account (`srcip_usersid`). If an account shares multiple files with outside entities, an alert is triggered. A sample Interflow includes the user ID (`srcip_usersid`), source IP address (`srcip`), and source country (`srcip_geo.countryName`).

Office 365 Malware Filter Policy Changed

The Microsoft Exchange malware filter policy changed. An overly permissive malware filter policy can allow attackers to send malicious attachments to your organization. Make sure this change was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Impair Defenses (T1562)
- Tags: [Office 365; Malware; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_malware_filter_policy_changed`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id
- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationId` — ID of the organization with the Microsoft Exchange malware policy change
- `OrganizationName` — organization with the Microsoft Exchange malware policy change

Use Case with Data Points

Office 365 monitors all Office 365 accounts (`srcip_usersid`) in every organization (`OrganizationId`) for Microsoft Exchange malware policy changes. If a change is discovered, an alert is triggered. A sample Interflow includes the account ID (`srcip_usersid`), organization ID (`OrganizationId`), and organization name (`OrganizationName`).

Office 365 Multiple Files Restored

Office 365 observed that multiple files were restored in a short period. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exploitation
- Tactic: Collection (TA0009)
- Technique: Data Staged (T1074)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_multi_file_restore`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `EventSource` — event source
- `srcip` — source IP address that caused the restore
- `srcip_host` — source host name

Use Case with Data Points

Office 365 periodically checks file restore records. If multiple file restore records are detected within a short period, an alert is triggered. A sample Interflow includes the Office 365 account ID (`srcip_usersid`), event source (`EventSource`), and source IP address (`srcip`).

Office 365 Multiple Users Deleted

Office 365 observed that multiple users were deleted in a short period. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Impact (TA0040)
- Technique: Account Access Removal (T1531)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_multi_user_deleted`.

Severity

50

Alert Suppression Fields 

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `EventSource` — event source
- `srcip` — source IP address that did the deletion
- `srcip_host` — source host name

Use Case with Data Points

Office 365 periodically checks user deletion records. If multiple users were deleted within a short period, an alert is triggered. A sample Interflow includes the Office 365 account ID (`srcip_usersid`), event source (`EventSource`), and source IP address (`srcip`).

Office 365 Network Security Configuration Changed

Office 365 identified a change to your organization's network security configuration, which is uncommon. Make sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Persistence (TA0003)
- Technique: Account Manipulation (T1098)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_security_conf_changed`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id
- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationId` — ID of the organization whose security configuration changed
- `OrganizationName` — name of the organization whose security configuration changed

Use Case with Data Points

Office 365 monitors all Office 365 accounts (`srcip_usersid`) in every organization (`OrganizationId`) for network security configuration changes. If a change is detected, an alert is triggered. A sample Interflow includes the Office 365 account ID (`srcip_usersid`), organization ID (`OrganizationId`), and organization name (`OrganizationName`).

Office 365 Password Policy Changed

Office 365 identified a change to the organization's password policy, which is uncommon. Make sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Modify Authentication Process (T1556)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_password_policy_changed`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id
- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationId` — ID of the organization whose password policy changed
- `OrganizationName` — name of the organization whose password policy changed

Use Case with Data Points

Office 365 monitors all Office 365 accounts (`srcip_usersid`) in every organization (`OrganizationId`) for sharing policy changes. If a change is detected, an alert is triggered. A sample Interflow includes the account ID (`srcip_usersid`), organization ID (`OrganizationId`), and organization name (`OrganizationName`).

Office 365 Sharing Policy Changed

Office 365 identified a change to your organization's sharing policy, which is uncommon. Make sure this was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Persistence (TA0003)
- Technique: Account Manipulation (T1098)
- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_sharing_policy_changed`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- service_id
- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationId` — ID of the organization whose sharing policy changed
- `OrganizationName` — name of the organization whose sharing policy changed

Use Case with Data Points

Office 365 monitors all Office 365 accounts (`srcip_usersid`) in every organization (`OrganizationId`) for password policy changes. If a change is detected, an alert is triggered. A sample Interflow includes the Office 365 account ID (`srcip_usersid`), organization ID (`OrganizationId`), and organization name (`OrganizationName`).

Office 365 User Network Admin Changed

The Office 365 account's network admin information was changed. Make sure this change was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Persistence (TA0003)
- Technique: Account Manipulation (T1098)

- Tags: [Office 365; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `office365_user_network_admin_changed`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the Office 365 account
- `OrganizationName` — name of the organization

Use Case with Data Points

Office 365 monitors the network admin information for each Office 365 account (`srcip_usersid`). If changes to the network admin are discovered, an alert is triggered. A sample Interflow includes the account ID (`srcip_usersid`) and organization name (`OrganizationName`).

Outbound Destination Country Anomaly

A host that typically communicates with a small, consistent number of countries communicated with a new country. Investigate the destination to see if it is benign.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Location Anomaly (XT2001)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `country_communication_anomaly`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip_geo.countryName` — name of the destination country
- `srcip` — source IP address
- `dstip` — destination IP address
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `appid_name` — application name
- `stability` — score measuring the time since the host communicated with the last new country
- `days_stable` — time since the host communicated with the last new country
- `diversity` — score measuring the number of countries with which the host communicated
- `child_count` — number of countries with which the host communicated

Use Case with Data Points

Hosts (`srcip_host`) and destination countries (`dstip_geo.countryName`) are examined periodically. If a host (`srcip_host`) with a small number of destination countries (`diversity`, `child_count`) has not visited a new country for a long time (`stability`, `days_stable`) visits a host (`dstip_host`) in a new country with an application (`appid_name`), an alert is triggered.

Outbytes Anomaly

A source IP address transmitted an anomalously high amount of outbound traffic. This could indicate data exfiltration.

This alert type has the following subtype:

- Outbytes Delta Anomaly



Firewall and non-firewall data do not contribute to the same alert, so this alert will have either entirely firewall data or no firewall data.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Exfiltration (TA0010)
- Technique: Automated Exfiltration (T1020)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `outbytes_anomaly`.

Severity

35

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_host 

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — source host name
- `actual` — actual amount of outbound traffic in the period
- `typical` — typical amount of outbound traffic from the source IP address
- `dstip_host` — destination host name

Use Case with Data Points

Every source host's (`srcip_host`) transferred data volume is calculated periodically. If a host's volume in outbytes_total (`actual`) is much higher than its normal volume (`typical`) in any period, an alert is triggered. The Interflow includes the destination host (`dstip_host`).

Alert Subtype: Outbytes Delta Anomaly 6.4.0

The Outbytes Delta Anomaly alert subtype is the same as the Outbytes Anomaly alert type above, with the following differences:

- The alert subtype is only available for vendors that support the outbytes_delta field.
- The description for this alert subtype is: A source IP address transmitted an anomalously high amount of outbound traffic. This momentary spike could indicate data exfiltration.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `outbytesdelta_spike`.
- The **Use Case with Data Points** is as follows:
  Every source host's (`srcip_host`) transferred data volume is calculated periodically. If a host's volume in outbytes_delta (`actual`) is much higher than its normal volume (`typical`) in any period, an alert is triggered. The Interflow includes the destination host (`dstip_host`).

Password Cracking with Hashcat

A user from a Windows host executed a command-line script that launched either the `hashcat.exe` command or a command using known Hashcat parameters (`-a -m 1000 -r`). The Hashcat command is known to use a SAM file from the Windows registry along with a password list to crack passwords.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Hashcat; Brute Force; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `password_cracking_with_hashcat`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip
- event_data.Image
- event_data.CommandLine

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — device internal IP address
- `computer_name` — name of the Windows host
- `event_data.CommandLine` — command used to run the tool
- `event_data.Image` — process running the hashcat tool

Use Case with Data Points

This alert is triggered if a Windows host (`hostip`) executes a PowerShell script with a context that includes one or more flags (`event_data.Image` or `event_data.CommandLine`) indicating usage of the Hashcat password cracking tool. The Interflow includes the IP address of the Windows host (`hostip`), the host name (`computer_name`), and the script image (`event_data.Image`) or script payload (`event_data.CommandLine`).

Validation / Remediation

Check the body of the Powershell script that is reported on the Windows host to identify whether the contents are actually malicious. If malicious, consider quarantining the host.

Potential False Positives

The running of any executable named `hashcat.exe` or any command that uses the hashcat signature parameter list (`-a -m 1000 -r`).

Password Resets Anomaly 5.2.0

An account reset/changed one or more target accounts' passwords for an anomalously large number of times. Check the subject account and major target accounts.

This alert type has the following subtype:

- Windows Account Password Reset Anomaly

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold

- Tactic: Persistence (TA0003)
- Technique: Account Manipulation (T1098)
- Tags: [Account Manipulation; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `password_resets_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_data.SubjectDomainName
- event_data.SubjectUserName

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_data.SubjectDomainName` — domain to which the `SubjectUserName` belongs
- `event_data.SubjectUserName` — user name of the account that resets/changes the password
- `actual` — actual time of the password resets/changes made by the user
- `typical` — expected maximum time of password resets/changes made by the user

Use Case with Data Points

The daily number of password reset/change actions of a user (`SubjectDomainName` + `SubjectUserName`) are monitored by (`actual`), which is compared with a dynamic upper threshold of (`typical`). An alert is triggered when the actual number exceeds the threshold.

Validation / Remediation

Validate the alert by checking the account activity on the date. If the number of resets/changes are abnormal, check the target user names that are being reset to verify if the action is expected.

Potential False Positives

False positives can be triggered in the following situations:

- Traffic pattern change, such as when an account is newly added or has some systematic change from the typical number of resets/changes
- Resets of usually silent accounts

Alert Subtype: Windows Account Password Reset Anomaly

The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `windows_account_password_resets_anomaly`.

Password Spraying Attempts Using Dsacls

A user from a Windows host executed a command-line script to launch a command that by name and parameter list indicates an attempt to abuse `dsacls.exe` for password spraying.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Signed Binary Proxy Execution (T1218)
- Tags: [Password Spray; Dsacls]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `password_spraying_attempts_using_dsacls`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name
- event_data.CommandLine
- event_data.OriginalFileName
- process_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — device internal IP address
- `computer_name` — name of the Windows host
- `event_data.CommandLine` — command used to run the tool
- `event_data.Image` — process running dsacls for password cracking
- `event_data.OriginalFileName` — actual file name that was executed

Use Case with Data Points

This alert is triggered if a Windows host (`hostip`) executes a `dsacls.exe` with a context that includes one or more flags (`event_data.Image`, `event_data.CommandLine`, or `event_data.OriginalFileName` including `/user` and `/passwd` as parameters). This indicates possible usage of Dcacls as a password spraying tool. The Interflow includes the IP address of the Windows host (`hostip`), the host name (`computer_name`), and the script image (`event_data.Image`) or the original file name (`event_data.OriginalFileName`), and script commandline (`event_data.CommandLine`).

Validation / Remediation

Check whether the usage was actually malicious. If so, consider quarantining the Windows host.

Potential False Positives

This alert could be triggered even if the use is a legitimate use of `dsacls` to bind to an LDAP session.

Phishing URL

A connection to a site with a phishing reputation was observed. Check with the user to determine whether their system is compromised.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: Initial Access (TA0001)
- Technique: Phishing (T1566)
- Tags: [Phishing; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `phishing`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- url
- dstip_host
- metadata.response.subject_alt_name
- username

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address of the connection to the phishing URL reputation site
- `dstip` — destination IP address of the phishing URL reputation site
- `url` — URL of the phishing site
- `dstip_host` — destination host name
- `metadata.response.subject_alt_name` — Subject Alternative Name of the phishing site
- `username` — name of the visitor
- `dstip_geo.countryName` — destination country
- `srcip_host` — source host name

Use Case with Data Points

If a connection from a source (`scrip`) to a site with a phishing reputation is detected, an alert is triggered. The Interflow includes the source IP address (`srcip`), source host (`srcip_host`), destination IP address (`dstip`), destination host (`dstip_host`), URL of the site (`url`), destination country (`dstip_geo.countryName`), Subject Alternative Name of the site (`metadata.response.subject_alt_name`), and user name (`username`).

Possible Encrypted Phishing Site Visit

A possible phishing site visit to a recently registered domain was observed in encrypted traffic. Check with the user to determine whether their system is compromised.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: Initial Access (TA0001)
- Technique: Phishing (T1566)
- Tags: [Phishing; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `encrypted_phishing_site`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.response.effective_tld` — effective top-level domain of the possible phishing site
- `srcip` — IP address of the visitor to the possible phishing site
- `dstip` — IP address of the possible phishing site
- `srcip_host` — source host name
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country

Use Case with Data Points

If an encrypted connection to a recently registered site (`metadata.response.effective_tld`) is observed, an alert is triggered. The Interflow includes the source IP address (`srcip`), source host (`srcip_host`), destination IP address (`dstip`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and effective top-level domain of the site (`metadata.response.effective_tld`).

Possible Phishing Site Visit from Email

A user visited a recently registered domain shortly after using email, indicating a possible phishing site visit. Check to see if the site is malicious. If so, check with the user to see if they are compromised.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: Initial Access (TA0001)
- Technique: Phishing (T1566)
- Tags: [Phishing; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `email_recent_domain_correlation`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip_domain_creation record's srcip, dstip, appid_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `recent_domain_id` — ID that points to the original record of the recently registered domain visit
- `email_traffic_id` — ID that points to the original record of email traffic
- `correlation_info.0.appid_name` — application on the visited domain accessed by the user
- `correlation_info.0.srcip` — IP address of the user
- `correlation_info.0.dstip` — IP address of the recently registered domain (useful if the `correlation_info.0.appid_name` **is not** DNS)
- `correlation_info.0.dstip_host` — recently registered domain that was visited (useful if the `correlation_info.0.appid_name` **is not** DNS)
- `correlation_info.0.metadata.response.query` — recently registered domain name the victim queried in DNS traffic. This field is only useful if the `correlation_info.0.appid_name` **is** DNS.
- `correlation_info.0.metadata.response.resolved_ips` — IP addresses of the recently registered domain name the victim queried in DNS traffic. This field is only useful if `correlation_info.0.appid_name` **is** DNS.

Use Case with Data Points

If a user (`srcip`) uses email (`appid_name`) and then either queries a recently registered (`metadata.response.domain_creation`) domain (`metadata.response.query`) or visits a recently registered (`dstip_domain_creation`) domain (`dstip_host`), an alert is triggered.

When an alert is triggered, a new correlation event is created. The Interflow includes the reference ID of the original record of the domain visit (`recent_domain_id`), the reference ID pointing to the original record of email traffic (`email_traffic_id`), the IP address of the user (`correlation_info.0.srcip`), the application involved in the recently registered site visit (`correlation_info.0.appid_name`), and the visited domain (`correlation_info.0.dstip_host` or `correlation_info.0.metadata.response.query`).

Logic Details of Possible Phishing Site Visit from Email

The timeframe of the alert type logic is as follows:

- The rule operates in cycles, running every 2 hours.
- During each run, it scans for email usage within the monitored traffic.
- After an email action, if a DNS lookup action occurs within a 30-minute window from the same source IP address, an alert is generated.

The domain query and DNS lookups are as follows:

- The domain creation date is queried from DNS traffic.
- The data points concerning domain creation dates are gathered from fields in the DNS traffic, including `dstip_domain_creation`, `metadata.response.domain_creation`, and `metadata.request.domain_creation`.
- The alert type logic considers a domain as *recent* if it has been created within the past 14 days.

The alert generation conditions are as follows:

- The alert is triggered if, within a 30-minute window, for the same source IP address, there is first, an email action, then there is DNS traffic indicating a domain creation event in the past 14 days from the same source IP address as the email's. (The rule runs every 2 hours, so this trigger could be delayed at most by 2 hours, but the two correlation events (email action and DNS traffic) need to happen within 30 minutes.)
- This logic ensures all specified conditions are met before an alert is generated.

Note that even though the rule runs every 2 hours, the 30-minute time window can be seen as a sliding window without gaps. For example: the rule runs now as a new cycle, but an email action happened during the last run, say 2 hours and 10 minutes ago. If the corresponding recent domain creation event happens in this cycle, say 1 hour and 50 minutes ago, this will also trigger the alert.

Possible Unencrypted Phishing Site Visit

A possible phishing site visit to a recently registered domain was observed in unencrypted traffic. Check with the user to determine whether their system is compromised.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts

- Tactic: Initial Access (TA0001)
- Technique: Phishing (T1566)
- Tags: [Phishing; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `unencrypted_phishing_site`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.response.effective_tld` — effective top-level domain of the possible phishing site
- `srcip` — IP address of the visitor to the phishing site
- `dstip` — IP address of the possible phishing site
- `srcip_host` — source host name
- `dstip_host` — destination host name
- `dstip_geo.countryName` — destination country

Use Case with Data Points

If an unencrypted connection to a recently registered site (`metadata.response.effective_tld`) is detected, an alert is triggered. The Interflow includes the source IP address (`srcip`), source host (`srcip_host`), destination IP address (`dstip`), destination host (`dstip_host`), destination country (`dstip_geo.countryName`), and effective top-level domain of the site (`metadata.response.effective_tld`).

PowerShell Remote Access

A Windows host executed a PowerShell script interacting with a remote host. Investigate the script and the remote host to determine whether the script is malicious. If so, consider quarantining the host.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Execution (TA0002)
- Technique: Command and Scripting Interpreter (T1059)
- Tags: [External; PowerShell]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `powershell_cnc`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name
- remote_ip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the Windows host
- `hostip_host` — host name
- `remote_ip` — IP address of the remote host involved in the script
- `event_data.ScriptBlockText` — contents of the PowerShell script

Use Case with Data Points

If a Windows host (`srcip`) executes a PowerShell script that includes potential communication (`event_data.ScriptBlockText`) with a remote host (`remote_ip`), an alert is triggered. The Interflow includes the IP address of the Windows host (`srcip`), the script body (`event_data.ScriptBlockText`), and the remote host IP address (`remote_ip`).

Rule Based Detection

The PowerShell Remote Access rules are used to identify when a Windows host executed a PowerShell script interacting with a remote host. Any one or more of these will trigger the PowerShell Remote Access alert type.

The rule-based detection is the same as the PowerShell Remote Access alert type above, with the following difference:

- The `detection_flag` is a Stellar Cyber enriched field:
  - **2200:** PowerShell script embedded with remote IP
  - **2201:** PowerShell script block with IP embedded at warning level (High fidelity)

Link to Rule-Based Alert Types

Rules Contributing to PowerShell Remote Access Alert Type

Private to Private Exploit Anomaly

A private IP address initiated a large number of exploit attempts identified by a given signature or a rarely seen exploit attempt to another private IP address. Investigate that signature.

This alert type has the following subtypes:

- IDS Traffic Anomaly
- IPS Traffic Anomaly

XDR Kill Chain

- Kill Chain Stage: Prop ion
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Exploitation of Remote Services (T1210)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `exploit_attempt_priv_priv`.

Severity

75

Alert Subtype: IDS Traffic Anomaly

The IDS Traffic Anomaly alert subtype is the same as the Private to Private Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ids_traffic_anomaly`.
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — signature of the exploit
- `ids.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IDS signatures (`ids.signature`), weighted by their severity (`ids.severity`), are calculated periodically. If many different exploits with unique IDS signatures are observed, an alert is triggered. . The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IDS signatures (`actual`), the usual accumulated severity

of IDS signatures (`typical`), and a sampling of the IDS signatures used in the attack (`ids_signatures_summarize`).

Alert Subtype: IPS Traffic Anomaly 5.2.0

The IPS Traffic Anomaly alert subtype is the same as the Private to Private Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ips_traffic_anomaly`.
- The Tags are: [Internal; Network Traffic Analysis; IPS Detection]
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ips.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ips.signature` — signature of the exploit
- `ips.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).

- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Private to Private IPS Signature Spike 5.2.0

A source IP address transmitted an anomalous number of different IPS signatures. Typically, this indicates host penetration or vulnerability scanning.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Lateral Movement (TA0008)
- Technique: Exploitation of Remote Services (T1210)
- Tags: [Internal; Network Traffic Analysis; IPS Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ips_signature_spike_priv_priv`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_summary.ips_signatures_summarize` — signatures of the exploit
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of unique IPS signatures in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of unique IPS signatures from the source IP address, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

Private to Public Exploit Anomaly

A private IP address initiated a large number of exploit attempts identified by a given signature or a rarely seen exploit attempt to a public IP address. Investigate that signature.

This alert type has the following subtypes:

- IDS Traffic Anomaly
- IPS Traffic Anomaly

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `exploit_attempt_priv_pub`.

Severity

60

Alert Subtype: IDS Traffic Anomaly

The IDS Traffic Anomaly alert subtype is the same as the Private to Public Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ids_traffic_anomaly`.
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — signature of the exploit

- `ids.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IDS signatures (`ids.signature`), weighted by their severity (`ids.severity`), are calculated periodically. If many different exploits with unique IDS signatures are observed, an alert is triggered. . The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IDS signatures (`actual`), the usual accumulated severity of IDS signatures (`typical`), and a sampling of the IDS signatures used in the attack (`ids_signatures_summarize`).

Alert Subtype: IPS Traffic Anomaly 5.2.0

The IPS Traffic Anomaly alert subtype is the same as the Private to Public Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ips_traffic_anomaly`.
- The Tags are: [External; Network Traffic Analysis; IPS Detection]
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ips.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ips.signature` — signature of the exploit
- `ips.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Private to Public IPS Signature Spike 5.2.0

A source IP address transmitted an anomalous number of different IPS signatures. Typically, this indicates host penetration or vulnerability scanning.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis; IPS Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ips_signature_spike_priv_pub`.

Severity

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_summary.ips_signatures_summarize` — signatures of the exploit
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of unique IPS signatures in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of unique IPS signatures from the source IP address, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

Process Anomaly

A process has been launched an anomalously large number of times. Investigate the process and the user to see if this is expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR Process Anomaly (XT1001)
- Tags: [Internal; Process Anomaly]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `bad_process`.

Severity

15

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- process_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name
- `process_name` — name of the process
- `actual` — actual number of launches in the period
- `typical` — typical number of launches in the period

Use Case with Data Points

The number of times a process (`process_name`) has been launched is calculated periodically. If the volume (`actual`) is much larger than the typical volume (`typical`) of the command or other commands in any period, an alert is triggered. The Interflow includes the (`hostip`) who launched the process.

Public to Private Exploit Anomaly

A public IP address initiated a large number of exploit attempts identified by a given signature or a rarely seen exploit attempt to a private IP address. Investigate that signature.

This alert type has the following subtypes:

- IDS Traffic Anomaly
- IPS Traffic Anomaly

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `exploit_attempt_pub_priv`.

Severity

60

Alert Subtype: IDS Traffic Anomaly

The IDS Traffic Anomaly alert subtype is the same as the Public to Private Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ids_traffic_anomaly`.
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — signature of the exploit
- `ids.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IDS signatures (`ids.signature`), weighted by their severity (`ids.severity`), are calculated periodically. If many different exploits with unique IDS signatures are observed, an alert is triggered. . The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IDS signatures (`actual`), the usual accumulated severity of IDS signatures (`typical`), and a sampling of the IDS signatures used in the attack (`ids_signatures_summarize`).

Alert Subtype: IPS Traffic Anomaly 5.2.0

The IPS Traffic Anomaly alert subtype is the same as the Public to Private Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ips_traffic_anomaly`.
- The Tags are: [External; Network Traffic Analysis; IPS Detection]
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ips.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ips.signature` — signature of the exploit
- `ips.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1

- `typical` — typical number of times this signature is seen in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

More on Internal versus External and Inbound vs Outbound...

Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Public to Private IPS Signature Spike 5.2.0

A source IP address transmitted an anomalous number of different IPS signatures. Typically, this indicates host penetration or vulnerability scanning.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis; IPS Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ips_signature_spike_pub_pr`

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_summary.ips_signatures_summarize` — signatures of the exploit
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of unique IPS signatures in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of unique IPS signatures from the source IP address, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

Public to Public Exploit Anomaly

A public IP address initiated a large number of exploit attempts identified by a given signature or a rarely seen exploit attempt to another public IP address. Investigate that signature.

This alert type has the following subtypes:

- IDS Traffic Anomaly
- IPS Traffic Anomaly

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `exploit_attempt_pub_pub`.

Severity



50

Alert Subtype: IDS Traffic Anomaly

The IDS Traffic Anomaly alert subtype is the same as the Public to Public Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ids_traffic_anomaly`.
- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ids.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ids.signature` — signature of the exploit
- `ids.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IDS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IDS signatures (`ids.signature`), weighted by their severity (`ids.severity`), are calculated periodically. If many different exploits with unique IDS signatures are observed, an alert is triggered. . The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IDS signatures (`actual`), the usual accumulated severity of IDS signatures (`typical`), and a sampling of the IDS signatures used in the attack (`ids_signatures_summarize`).

Alert Subtype: IPS Traffic Anomaly 5.2.0

The IPS Traffic Anomaly alert subtype is the same as the Public to Public Exploit Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `ips_traffic_anomaly`.
- The Tags are: [External; Network Traffic Analysis; IPS Detection]

- The **Alert Suppression Fields**, **Key Fields and Relevant Data Points**, and **Use Case with Data Points** are as follows:

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- ips.signature

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `ips.signature` — signature of the exploit
- `ips.severity` — severity of the exploit
- `actual` — actual number of times this signature was found in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of times this signature is seen in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:
>
> - Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
> - Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
> - When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
> - Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction

of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Public to Public IPS Signature Spike 5.2.0

A source IP address transmitted an anomalous number of different IPS signatures. Typically, this indicates host penetration or vulnerability scanning.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] Initial Access (TA0001)
- Technique: Exploit Public-Facing Application (T1190)
- Tags: [External; Network Traffic Analysis; IPS Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ips_signature_spike_pub_pub`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_summary.ips_signatures_summarize` — signatures of the exploit
- `srcip_host` — host name of corresponding source IP address
- `actual` — actual number of unique IPS signatures in the period, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1
- `typical` — typical number of unique IPS signatures from the source IP address, with *critical* IPS signatures counted as 2, *high* counted as 1, *low* counted as 0.5, and others counted as 1

Use Case with Data Points

The number of unique IPS signatures (`ips.signature`), weighted by their severity (`ips.severity`), are calculated periodically. If many different exploits with unique IPS signatures are observed, an alert is triggered. Additionally, the action (`ips.action`) taken by the IPS affects the alert fidelity. The Interflow includes a source IP address (`srcip`), timestamp, an accumulated severity of IPS signatures (`actual`), the usual accumulated severity of IPS signatures (`typical`), and a sampling of the IPS signatures used in the attack (`ips_signatures_summarize`).

RDP Outbytes Anomaly

An internal host transferred an anomalously high amount of data to external host(s) through RDP. This could indicate data exfiltration. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Exfiltration (TA0010)
- Technique: Exfiltration Over Alternative Protocol (T1048)
- Tags: [RDP; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_outbytes_anomaly`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — destination IP address
- `dstip_host` — destination host name
- `actual` — actual amount of outbound traffic in the period
- `typical` — typical amount of outbound traffic from the destination IP address
- `srcip_host` — source IP address that initiates the RDP connection

Use Case with Data Points

Every destination host's (`dstip`) transferred data volume through RDP is calculated periodically. If a host's volume (`actual`) is much greater than normal (`typical`) in any period, an alert is triggered. A sample Interflow includes the destination host (`dstip_host`).

RDP Port Opening

Netsh commands to open TCP port 3389 were observed. This could indicate Sarwent malware attempting to establish an RDP connection. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Impair Defenses (T1562)
- Tags: [RDP]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_port_opening`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — source IP address that executes the command
- `hostip_host` — host name
- `process_name` — process name
- `event_data.CommandLine` — command line used

Use Case with Data Points

Commands that open TCP port 3389 are monitored, and if netsh commands are seen, an alert is triggered. A sample Interflow includes the source IP address (`hostip`) and the command used (`event_data.CommandLine`).

RDP Registry Modification

Modifications of the property values of `fDenyTSConnections` and `UserAuthentication` to enable remote desktop connections were observed. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Modify Registry (T1112)
- Tags: [RDP]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_registry_modification`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host that made the setting change
- `hostip_host` — host name
- `event_data.TargetObject` — name of the registry key
- `event_data.Details` — value of the registry

Use Case with Data Points

The property values of `fDenyTSConnections` and `UserAuthentication` are monitored, and if a possible malicious modification of the settings to enable remote desktop connections is observed, an alert is triggered. A sample Interflow includes the source IP address (`hostip`) and the registry name (`event_data.TargetObject`).

RDP Reverse Tunnel

An `svchost` hosting RDP `termsvcs` communicating with the loopback address on TCP port 3389 was observed. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Command and Control (TA0011)
- Technique: Protocol Tunneling (T1572)
- Tags: [RDP]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_reverse_tunnel`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name
- `event_data.Image` — process communicating with the loopback address

Use Case with Data Points

If an svchost hosting RDP termsvcs communicating with the loopback address is found on TCP port 3389, an alert is triggered. A sample Interflow includes the host IP address (`hostip`) and host name (`hostip_host`).

RDP Session Hijacking

A suspicious RDP session using `tscon.exe` or MSTSC shadowing was observed. This could indicate a hijacked RDP session. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: Lateral Movement (TA0008)
- Technique: Remote Service Session Hijacking (T1563)
- Tags: [RDP; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_session_hijacking`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address that executes the command
- `hostip_host` — host name
- `process_name` — process name
- `event_data.CommandLine` — command line used

Use Case with Data Points

If an RDP session redirect using `tscon.exe` or MSTSC is detected, an alert is triggered. A sample Interflow includes the host IP address (`hostip`), name of the process used (`process_name`), and command used (`event_data.CommandLine`).

RDP Settings Hijacking

Changes to RDP terminal services settings were observed. Check the IP address and block if necessary.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Defense Evasion (TA0005)
- Technique: Modify Registry (T1112)
- Tags: [RDP]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_settings_hijack`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host that made the setting change
- `hostip_host` — host name
- `event_data.TargetObject` — name of the registry key
- `event_data.EventType` — event type on the registry key (SetValue, DeleteValue)
- `event_data.Details` — value of the registry

Use Case with Data Points

RDP terminal service settings are monitored, and if changes are found to these settings, an alert is triggered. A sample Interflow includes the source IP address (`hostip`) and the registry name (`event_data.TargetObject`).

RDP Suspicious Logon

An RDP logon with a local source IP address was observed. This could indicate a tunneled logon. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Command and Control (TA0011)
- Technique: Protocol Tunneling (T1572)
- Tags: [RDP]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_suspicious_logon`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address of the RDP server
- `event_data.TargetDomainName` — domain of the login account
- `event_data.TargetUserName` — user name of the login account
- `hostip_host` — host name

Use Case with Data Points

Remote desktop logins are monitored, and if a local source IP address is seen, an alert is triggered. A sample Interflow includes the source IP address (`hostip`) and host name (`hostip_host`).

RDP Suspicious Logon Attempt

An authenticated user who is not allowed to log on remotely has attempted to connect through RDP. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Internal; RDP; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `rdp_suspicious_logon_attempt`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address of the RDP server
- `hostip_host` — host name
- `event_data.AccountDomain` — account domain of the user who attempts to connect
- `event_data.AccountName` — account name of the user who attempts to connect
- `event_data.ClientAddress` — IP address of the user who attempts to connect

Use Case with Data Points

Windows remote desktop logins are monitored, and if a user who is not allowed to remotely log in tries to log in with RDP, an alert is triggered. A sample Interflow includes the source IP address

(`hostip`) and host name (`hostip_host`).

Recently Registered Domains

A DNS request was observed for a site that was registered less than 90 days ago. Check the domain. If suspicious, notify users.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR New Domain (XT2008)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `new_registered_domain`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.effective_tld
- dns.question.registered_domain

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `metadata.request.effective_tld` — top-level domain name in the request
- `srcip_host` — host name of corresponding source IP address
- `dstip_host` — host name of corresponding destination IP address
- `metadata.response.domain_creation` — domain creation time
- `metadata.response.effective_tld` — top-level domain name in the response
- `metadata.response.resolved_ips` — list of resolved IP addresses
- `actual` — number of visits to the domain in the period
- `domain_creation` — domain creation time 5.2.0 5.3.0
- `dns.question.registered_domain` — highest registered domain 5.2.0 5.3.0
- `dns.question.name` — domain name in request 5.2.0 5.3.0

Use Case with Data Points

If a domain has been registered within the last 90 days, an alert is triggered. A sample Interflow includes the domain name (`metadata.request.effective_tld`), source host (`srcip_host`), destination host (`dstip_host`), and domain creation time (`metadata.response.domain_creation`).

Remote Access Pattern Anomaly 6.2.0

An uncommon pattern of applic ns usage was observed. Investigate if the source host has unexpected application connections. It could be a malware infected host.

This alert type focuses on suspicious file-sharing/storage apps (onedrive, quic, github, wireguard, and openvpn).

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: [External] Exfiltration (TA0010)
- Technique: Exfiltration Over Web Service (T1567)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `external_appflow_outlier`.

Severity

40

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- appid_name
- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `appid_name` — name of the remote access application
- `dstip_type` — IP address type of the remote access destination
- `num_clust` — number of remote access patterns
- `sum_clust_size` — number of source IP addresses that have a similar remote access pattern
- `training_days` — number of days of remote access data

Use Case with Data Points

The usage pattern from (`srcip`) of (`dstip_type`) (`appid_name`) is unusual compared to other (`num_clust`) different types of usage patterns observed from (`sum_clust_size`) source IP addresses in (`training_days`) days.

> *More on Internal versus External and Inbound vs Outbound...*
>
> Stellar Cyber reports both *internal* and *external* versions of some alerts, with different analysis and recommended actions for each. Similarly, IDS signatures report the direction of data flow as `inbound` or `outbound`. Use the following as a guide for these concepts:

- Addresses with a `srcip_type` and `dstip_type` of `private` are identified as **internal**. All other values are identified as **external** (when applicable; not all alerts have unique analytics for internal/external).
- Communications between hosts where `srcip_type` and `dstip_type` are both `private` are considered **internal** communications.
- When an anomaly is observed on an **internal communication**, the attack is considered to be **internal**.
- Stellar Cyber always sets the `srcip` in the Interflow record as the initiating IP address of an event. Note that IDS signatures, which are reported with relevant alerts, instead list addresses based on the direction of data flow, not the initiating address. So an INBOUND data flow may show the `dstip` as the source address and the `srcip` as the destination address, even though the `srcip` was the initiator of the request. Use INBOUND and OUTBOUND information in the signature to understand the direction of data flow, together with Stellar Cyber's Interflow record of `srcip` and `dstip` to understand which address initiated the threat event.

Scanner Reputation Anomaly

An anomalously large amount of connections were observed from an IP address with a reputation of being a scanner. Cross-check with the **IP / Port Scan Anomaly** alert, and check the links and content for possible spam or phishing.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: Reconnaissance (TA0043)
- Technique: Active Scanning (T1595)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `scanner_rep`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_host

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — host name of corresponding source IP address
- `srcip_reputation` — source reputation
- `srcip_geo.countryName` — source country
- `actual` — actual number of connections from this source in the period
- `typical` — typical number of connections from this source in the period

- `dstip_host` — host name of corresponding destination IP address

Use Case with Data Points

The number of connections from a source IP address (`srcip_host`) with a reputation as a scanner (`srcip_reputation`) is calculated every 5 minutes. If the number of connections (`actual`) is much greater than normal (`typical`), an alert is triggered. The Interflow includes information such as the source country (`srcip_geo.countryName`) and a destination (`dstip_host`).

Sensor Status Anomaly

The sensor has changed its status from "connected" to "disconnected".

For Windows Agent sensors, this alert type logs the status of each sensor every 5 minutes. The status includes whether or not the sensor is connected and whether or not the sensor is sending data. 5.3.0

This alert type has the following subtypes: 5.3.0

- Sensor Status Flipped
- Disconnected but still sending data
- Disconnected and stopped sending data
- Disconnected and inactive

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: XDR SBA (XTA0003)
- Technique: XDR Status Anomaly (XT3002)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ade_outbytes_anomaly_flip`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- engid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `engid` — sensor ID
- `engid_name` — sensor name
- `metadata.status` — sensor (engine) status 5.3.0

Use Case with Data Points

For each sensor, its connection status is checked periodically, if the status changes from “connected“ to “disconnected“, the anomaly is triggered. A sample Interflow includes the sensor ID (`engid`) and sensor name (`engid_name`).

Alert Subtype: Sensor Status Flipped 5.3.0

The Sensor Status Flipped alert subtype is the same as the Sensor Status Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `sensor_status_anomaly_flip`.
- The subtype is for data sources from Windows Agent sensors.
- It is triggered instantly when a disconnection is logged.

Alert Subtype: Disconnected but still sending data 5.3.0

The Disconnected but still sending data alert subtype is the same as the Sensor Status Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `sensor_status_anomaly_sending_data`.
- The subtype is for data sources from Windows Agent sensors.
- It indicates a connection status of **disconnected but still sending data**.
- A disconnection alert is sent if the sensor stays disconnected or sends no data for 15 minutes.

Alert Subtype: Disconnected and stopped sending data 5.3.0

The Disconnected and stopped sending data alert subtype is the same as the Sensor Status Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `sensor_status_anomaly_stopped_sending_data`.
- The subtype is for data sources from Windows Agent sensors.
- It indicates a connection status of **disconnected and stopped sending data**.
- A disconnection alert is sent if the sensor stays disconnected or sends no data for 15 minutes.

Alert Subtype: Disconnected and inactive 5.3.0

The Disconnected and inactive alert subtype is the same as the Sensor Status Anomaly alert type above, with the following differences:

- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `sensor_status_anomaly_no_data`.
- The subtype is for data sources from Windows Agent sensors.
- It indicates a transition from a connection status of **connected, not sending data** to **disconnected, not sending data**.

- A disconnection alert is sent if the sensor stays disconnected or sends no data for 15 minutes.

SMB Impacket Lateralization

The execution of `wmiexec, dcomexec, atexec, smbexec, or PSExec` from the Impacket framework was observed. Check the source host. If malicious, consider blocking the host.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Execution (TA0002)
- Technique: Windows Management Instrumentation (T1047)
- Tags: [SMB; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `smb_impacket_lateralization`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name
- `event_data.CommandLine` — command line of the command that was executed
- `event_data.ParentCommandLine` — command line of the parent process

Use Case with Data Points

If a Windows host (`srcip`) executes a command (`wmiexec, dcomexec, atexec, smbexec`, or `PSExec`) from the Impacket framework, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), and the command executed (`event_data.CommandLine`).

SMB Specific Service Installation

A specific service installation used by the Impacket tool or Metasploit was observed. Check the source host. If malicious, consider blocking the host.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: Execution (TA0002)

- Technique: System Services (T1569)
- Tags: [SMB]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `smb_hack_smbexec`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- event_data.ServiceName

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name
- `event_data.ServiceName` — name of the service installed

Use Case with Data Points

If a Windows host (`srcip`) installs a specific service installation that is used by the `smbexec.py` tool, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), and the service installed (`event_data.ServiceName`).

SMB Suspicious Copy

A suspicious copy command from a remote C$ or ADMIN$ share was observed. Check the source host. If malicious, consider blocking the host.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: Collection (TA0009)
- Technique: Data from Network Shared Drive (T1039)
- Tags: [SMB]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `smb_suspicious_copy`.

Severity

75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — host IP address
- `hostip_host` — host name
- `event_data.CommandLine` — command line of the copy command

Use Case with Data Points

If a Windows host (`srcip`) uses the copy command to copy files from a remote C$ or ADMIN$ share, an alert is triggered. A sample Interflow includes the source IP address (`srcip`), source host (`srcip_host`), and the command executed (`event_data.CommandLine`).

Successful Login After Brute Force 6.5.0

A successful login occurred from a new IP address after the user's account experienced brute force activity. Verify the login is legitimate and investigate the source of the failed attempts.

You can customize the Successful Login After Brute Force alert type in the Detection Management System. See Configuring Customizations for ML Detections.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: Credential Access (TA0006)
- Technique: Brute Force (T1110)
- Tags: [Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `brute_force_success`.

Severity

90

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — user account that experienced the brute force activity
- `srcip` — source IP address of the successful login
- `login_type` — type of login (for exmaple, interactive, network)

- `event_summary.unique_failure_ips_short` — number of unique IP addresses with login failures in the last 30 minutes
- `event_summary.unique_failure_ips_long` — number of unique IP addresses with login failures in the last 24 hours
- `event_summary.failure_ips` — list of IP addresses that had failed login attempts (up to 100)
- `srcip_asn` — Autonomous System Number (ASN) of the source IP address of the successful login that triggered the alert
- `srcip_asn_org` — autonomous system organization of the source IP address of the successful login that triggered the alert
- `srcip_geo` — source IP address geo location of login success, including latitude and longitude
- `srcip_host` — hostname associated with the source IP address of the successful login that triggered the alert
- `srcip_reputation` — reputation of the source IP address of the successful login that triggered the alert
- `event_summary.short_window_triggered` — whether the 30-minute failure window threshold was triggered
- `event_summary.long_window_triggered` — whether the 24-hour failure window threshold was triggered

Use Case with Data Points

This alert type is triggered if a user (`srcip_usersid`) has login failures from multiple different IP addresses within a 30-minute or 24-hour window followed by a login success from an IP address with no recent login successes. The number of login failures required for each window can be customized in the Detection Management System.

The login success is the basis for the Interflow. It will be enriched with information about the login failures. The `event_summary.failure_ips` field lists the IP addresses with failures. The `unique_failure_ips_short` and `unique_failure_ips_long` fields list the number of unique failure IP addresses within the 30-minute and 24-hour windows respectively.

Suspected AS-REP Roasting Attacks 6.2.0

Suspected Authentication Server Response (AS-REP) Roasting attacks have been observed. These are suspicious Kerberos authentication events without pre-authentication. Investigate the source IP address and target user name.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] Credential Access (TA0006)
- Technique: Steal or Forge Kerberos Tickets (T1558)
- Sub-technique: AS-REP Roasting (T1558.004)
- Tags: [Internal; Kerberos; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `kerberos_roasting`.

Severity



75

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `login_result` — login result
- `event_data.PreAuthType` — pre-authentication type (0: disabled)
- `event_summary.kerberos_roasting.user_summary` — user names without pre-authentication
- `actual` — actual number of users without pre-authentication in 5-minute interval
- `threshold` — typical number of users without pre-authentication in 5-minute interval

Use Case with Data Points

Detect AS-REP roasting by checking for spikes in the number of Kerberos Ticket Granting Ticket (TGT) requests without pre-authorization.

An unusually high number (`actual`) of user accounts without pre-authentication was observed from the source IP address (`srcip`). The expected number of user accounts without pre-authentication for (`srcip`) is (`threshold`), measured in a 5-minute interval. This indicates a potential AS-REP roasting attack.

Suspected Network Beaconing Activities 6.1.0

Suspected network beaconing activity was observed. If malicious, block the IP address of the Command and Control server and consider containing the source hosts.

Command and Control (C&C or C2) beaconing describes the behavior of bots actively, persistently, and regularly connecting to a C2 server. (The regular interval is a characteristic allowing detection.)

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: [External] XDR Intel (XTA0005)
- Technique: Application Layer Protocol (T1071)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `c2beaconing`.

Severity

70

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip
- dstport

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `dstip` — destination IP address
- `srcip` — source IP address
- `period_cnt` — number of beaconing connections observed
- `period_mean` — frequency of beaconing connections
- `dstport` — destination port
- `connection_prevalence` — prevalence of beaconing connection destination

Use Case with Data Points

The beaconing behavior between source host (`srcip`) and destination (`dstip`):(`dstport`) has frequency and persistency. The c2beaconing frequency (`period_mean`) should be between 100 to 400 seconds and the behavior should be persistent (`period_cnt`) for over a day. A low prevalence connection means the destination IP address belongs to an ASN to which only a few hosts connect. If a connection is low prevalence and the beaconing behavior has frequency and persistency, an alert is triggered.

Suspicious LDAP Search Request 5.5.0

A suspicious LDAP search request was observed for the first time. This behavior indicates that an attacker might be attempting to collect information from the Active Directory (AD) service.

To turn on structured LDAP query parsing, enable `ldap-metadata` in the sensor CLI.

This alert type has the following subtypes:

- Domain Trust Enumeration
- All Object Enumeration
- Wildcard Query

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: Discovery (TA0007)
- Technique: Account Discovery (T1087)
- Tags: [LDAP; Network Traffic Analysis; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `suspicious_ldap`.

Severity

74

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip
- dstip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_host` — source host that sent the suspicious LDAP search request
- `srcip_geo.countryName` — country of the suspicious source host
- `dstip_host` — destination host to which the suspicious LDAP search request was sent
- `metadata.request.base_object` — baseObject parameter of the suspicious LDAP search request
- `metadata.request.scope` — scope parameter of the suspicious LDAP search request
- `metadata.request.filter` — filter parameter of the suspicious LDAP search request

Use Case with Data Points

The parameters (`metadata.request.filter`) of LDAP requests captured from network traffic are examined. If an LDAP request with suspicious parameters is seen for the first time, an alert is triggered. Each alert subtype has its own criteria for detecting suspicious parameters. See the descriptions below.

Validation / Remediation

Check the `metadata` field of the alert to see if the LDAP requests are legitimate. If the `metadata` looks suspicious, further investigate the source IP address that is sending the requests.

Potential False Positives

LDAP is widely used in AD environments by legitimate services to exchange domain information. If the user performs a domain backup or DCSync operation, the corresponding LDAP traffic may be flagged as suspicious.

Alert Subtype: Domain Trust Enumeration

The Domain Trust Enumeration alert subtype is the same as the Suspicious LDAP Search Request alert type above, with the following differences:

- The alert subtype detects domain trust enumeration behaviors from LDAP request parameters. The subtype is for LDAP search requests that have scope set to 2 (`wholeSubtree`) and filter set to (`objectclass=trusteddomain`).
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `suspicious_ldap_domaintrust_enumeration`.

- Technique: Domain Trust Discovery (T1482).

Alert Subtype: All Object Enumeration

The All Object Enumeration alert subtype is the same as the Suspicious LDAP Search Request alert type above, with the following differences:

- The alert subtype detects all object enumeration behaviors from LDAP request parameters. The subtype is for LDAP search requests that have scope set to 2 (`wholeSubtree`) and filter set to (`objectclass=*`).
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `suspicious_ldap_all_object_enumeration`.
- Technique: Account Discovery (T1087).

Alert Subtype: Wildcard Query

The Wildcard Query alert subtype is the same as the Suspicious LDAP Search Request alert type above, with the following differences:

- The alert subtype detects wildcard queries in LDAP requests. The subtype is for LDAP search requests that have an asterisk (*) in the filter part, excluding some false positives.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `suspicious_ldap_wildcard_query`.
- Technique: Account Discovery (T1087).
- Has an Alert Suppression Field, metadata.request.filter, in addition to srcip and dstip.

Suspicious OCI Cross Tenancy Communication 6.1.0

This topic covers a feature that is not available for all customers yet. See Early Access Program Features and Topics Under Development.

An outbound request from an OCI tenant targeting another tenancy is observed, signaling potential lateral movement.

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: Lateral Movement (TA0008)
- Technique: Remote Services (T1021)
- Sub-technique: Direct Cloud VM Connections (T1021.008)
- Tags: [OCI]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `suspicious_oci_cross_tenancy_communication`.

Severity

25

Alert Suppression Fields 

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- oracle.oracle.tenantid
- oracle.data.identity.tenantId

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `oracle.oracle.tenantid` — tenancy owning the resource targeted by the event
- `oracle.data.identity.tenantId` — tenancy emitting the event (the tenant from which the request was made)

Use Case with Data Points

Stellar Cyber detects OCI cross-tenancy communication (that is, outbound requests from a tenant targeting another tenancy) by checking whether the tenant from which the request was made (`data.identity.tenantId`) is identical to the tenancy that owns the resource targeted by the request (`oracle.tenantId`). Note that `data.identity.tenantId` does not always exist (for example, if an action was performed by Oracle-managed control plan services and therefore did not originate from an end-user or resource with a tenant association), but `oracle.tenantId` should always exist.

Unapproved Asset Activity

Activity of an asset that was marked as unapproved in one of the Asset Analytics tabs has been observed.

Unapproved assets generate one alert per day until their approval status is changed with either the **Approve** or **Ignore** button in the Asset Analytics tabs.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR Unapproved Asset Activity (XT2013)
- Tags: [Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `asset_violation`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- asset_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `asset_id` — ID of a specific asset
- `vendor` — vendor of this asset

Use Case with Data Points

If an analyst marks an asset as unapproved in the **Asset Activity** tabs, a daily alert is triggered until the asset is either manually approved or ignored.

Uncommon Application Anomaly

Private (internal assets) to public (Internet) traffic has revealed an application that has never been seen before (or been seen very rarely). Investigate that application and ensure that it is benign.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR App Anomaly (XT2003)
- Tags: [External; Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `network_uncommon_app`.

Severity

20

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- appid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `appid_name` — application name
- `days_silent` — number of days since this application was last seen
- `srcip_host` — host name of corresponding source IP address
- `srcip_reputation` — source reputation
- `srcip_geo.countryName` — source country
- `dstip_host` — host name of corresponding destination IP address
- `dstip_reputation` — destination reputation
- `dstip_geo.countryName` — destination country

Use Case with Data Points

If an application (`appid`) has never been observed by Stellar Cyber or been seen very rarely (`days_silent`), an alert is triggered. The Interflow includes the internal assets (`srcip_host`), source reputation (`srcip_reputation`), and source country (`srcip_geo.countryName`), and the destination host (`dstip_host`), destination reputation (`dstip_reputation`), and destination country (`dstip_geo.countryName`).

Uncommon Process Anomaly

An asset launched a process that has never been seen before (or has very rarely been seen). This could indicate a malware attack.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR Process Anomaly (XT1001)
- Tags: [Process Anomaly]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `network_uncommon_process`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- process_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host running the process
- `hostip_host` — host name
- `process_name` — name of the process
- `wineventlog_user` — user that created the process
- `days_silent` — number of days since this process was last seen

Use Case with Data Points

If a process (`process_name`) has never been observed by Stellar Cyber or been seen very rarely (`days_silent`), an alert is triggered. The Interflow includes the user (`process_user`) and host (`srcip`) that executed the process.

Uncommon Top-Level Domain Anomaly 6.1.0

A host has sent a suspicious DNS query with a top-level domain (TLD) that has never been seen before (or has very rarely been seen). This could indicate a malware attack. Check the DNS domain and consider blocking the host.

New or unusual TLDs might signal a threat vector, for example:

- Threat actors often register new domains with rare or newly delegated TLDs to avoid blacklists (such as, .zip, .click, or .lol).
- These TLDs are frequently used for malware delivery, phishing, or Command and Control (C&C or C2) infrastructure, especially in the early days before domain reputation systems catch up.
- Detecting new TLDs can surface early-stage recon or delivery attempts before other threat intelligence flags them.



The data sources for this alert type are Stellar Cyber sensor and third-party data from vendors such as HYAS.

This alert type has the following subtypes:

- Sensor DNS Data Source
- Vendor DNS Data Source

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: Command and Control (TA0011)
- Technique: Application Layer Protocol (T1071)
- Tags: [DNS Anomaly; Top-Level Domain]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `dns_new_tld`.

Severity

20

Alert Subtype: Sensor DNS Data Source

The xdr_event.subtype.name for this alert subtype in the Interflow data is `dns_new_tld_sensor`.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.query
- metadata.request.effective_tld

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — source host name
- `metadata.request.query` — DNS query domain, such as www.stellarcyber.ai

- `metadata.request.effective_tld` — effective top-level domain, such as stellarcyber.ai
- `days_silent` — number of days since last activity

Use Case with Data Points

If a DNS query is observed to have a new top-level domain, an alert is triggered. The Interflow includes a DNS query domain (`metadata.request.query`), a top-level domain (`metadata.request.effective_tld`), the number of days since the same TLD was seen last time (`days_silent`), and the last seen timestamp for the TLD (`last_seen_ts`).

Validation / Remediation

Validate the alert by checking the suspiciousness of the DNS query and block the query domain if the domain is suspicious.

Potential False Positives

- Legitimate software updates may trigger unusual DNS queries as they contact uncommon domains for downloading updates.
- Internal applications using dynamic DNS services might generate rare DNS queries.
- Third-party security tools or monitoring solutions may use unique DNS queries for their operations
- Cloud services often use diverse and uncommon domains for legitimate operations.
- New or infrequently accessed legitimate websites may appear as unusual.

Alert Subtype: Vendor DNS Data Source

The xdr_event.subtype.name for this alert subtype in the Interflow data is `dns_new_tld_vendor`.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- dns.question.name
- dns.question.top_level_domain

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — source host name
- `dns.question.name` — DNS query domain, such as www.stellarcyber.ai
- `dns.question.top_level_domain` — normalized top-level domain, such as .ai
- `days_silent` — number of days since last activity

Use Case with Data Points

If a DNS query is observed to have a new top-level domain, an alert is triggered. The Interflow includes a DNS query domain (`dns.question.name`), a top-level domain (`dns.question.top_level_domain`), the number of days since the same TLD was seen last time (`days_silent`), and the last seen timestamp for the TLD (`last_seen_ts`).

Validation / Remediation

Validate the alert by checking if  DNS query is suspicious and block the query domain if the domain is suspicious.

Potential False Positives

- Legitimate software updates may trigger unusual DNS queries as they contact uncommon domains for downloading updates.
- Internal applications using dynamic DNS services might generate rare DNS queries.
- Third-party security tools or monitoring solutions may use unique DNS queries for their operations
- Cloud services often use diverse and uncommon domains for legitimate operations.
- New or infrequently accessed legitimate websites may appear as unusual.

User Asset Access Anomaly

A user who typically uses a small, consistent number of assets was observed in Windows logons, Kerberos service tickets, or SMB traffic to access a new asset. Investigate the asset and user to see if this was expected.

This alert type has the following subtypes:

- SMB User Based
- Kerberos TGS Requests

XDR Kill Chain

- Kill Chain Stage: Propagation
- Tactic: [Internal] XDR UBA (XTA0004)
- Technique: XDR Asset Anomaly (XT4004)
- Tags: [Internal; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `user_asset_access`.

Severity

30

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_data.TargetUserName
- metadata.request.username
- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — source user ID
- `dstip_host` — host name of corresponding destination IP address
- `srcip_host` — host name of corresponding source IP address
- `srcip_username` — source user name
- `stability` — score measuring the time since the last new asset was accessed
- `days_stable` — time since the last new asset was accessed
- `diversity` — score measuring the number of assets that the user accessed
- `child_count` — number of assets that the user accessed

Use Case with Data Points

Users (`srcip_usersid` and `srcip_username`) with a small number of assets (`diversity`, `child_count`) who also have not used a new asset (`srcip_host`) for a long time (`stability`, `days_stable`) are examined. If a new asset appears on a host (`srcip_host`) with this user, an alert is triggered.

The user is identified with the `scrip_userid` and `scrip_username` fields. The asset is identified with the `scrip_host` field. Active Directory, which is identified from the `dstip_host` field, provides the relationship between the user and the asset. Stability is identified with the `stability` field and diversity is identified with the `diversity` field.

Alert Subtype: SMB User Based 5.2.0

The SMB User Based alert subtype is the same as the User Asset Access Anomaly alert type above, with the following differences:

- The subtype is more specific to SMB users authenticating to a new asset. It uses network traffic to monitor the network shares the users accessed.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `smb_user_asset_access`.
- It has the following **Alert Suppression Fields** and **Key Fields and Relevant Data Points**.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- metadata.request.username

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `dstip` — destination IP address
- `event_summary.ueba_smb_username` — SMB user that accessed the assets
- `stability` — score measuring the time since the last new asset was accessed
- `days_stable` — time since the last new asset was accessed
- `diversity` — score measuring the number of assets that the user accessed
- `child_count` — number of assets that the user accessed

Alert Subtype: Kerberos TGS Requests 5.4.0

The Kerberos TGS Requests alert subtype is the same as the User Asset Access Anomaly alert type above, with the following differences:

- The subtype enhances detection of anomalous user asset relations on Windows environments by monitoring Kerberos Ticket Granting Service (TGS) requests.
- The `xdr_event.subtype.name` for this alert subtype in the Interflow data is `kerberos_user_asset_access`.
- It has the following **Alert Suppression Fields** and **Key Fields and Relevant Data Points**.

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event_data.TargetUserName

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event_data.IPAddress` — IP address
- `event_data.ServiceName` — Service Principal Name (SPN) servicename
- `event_data.TargetUserName` — SPN user name
- `stability` — score measuring the time since the last new asset was accessed
- `days_stable` — time since the last new asset was accessed
- `diversity` — score measuring the number of assets that the user accessed
- `child_count` — number of assets that the user accessed

User Login Location Anomaly

A login to a user account occurred from a source IP address that is anomalously distant from the nearest location typically observed for logins to that user account.

This alert type has a detection delay for on-time records while maintaining detection coverage for high latency data sources. High latency data will have a detection delay corresponding to their amount of latency.

The expected detection delay is 5-10 minutes, although it could be longer when there is an ingestion delay. Sources without ingestion delays will get their alerts between 5 and 10 minutes after ingestion.

You can customize the User Login Location Anomaly in the Detection Management System. See Configuring Customizations for ML Detections.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR UBA (XTA0004)
- Technique: XDR Location Anomaly (XT2001)
- Tags: [External; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `user_login_region`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — key ID for the source user
- `distance_deviation` — deviation in distance between two login locations (miles)
- `srcip_host` — source host name
- `srcip_reputation` — reputation of the source IP address
- `srcip_geo.countryName` — source IP country name
- `srcip_geo.region` — source IP region name
- `srcip_geo.city` — source IP city name
- `login_type` — type of login
- `has_mfa` — whether a Multi-Factor Authentication (MFA) event is associated with the current login 5.5.0
- `appId` — application ID 6.5.0
- `appDisplayName` — application display name 6.5.0

Use Case with Data Points

Successful login events for certain login types (`login_type`) of a user (`srcip_usersid`) from a source host (`srcip_host`) and country location (`srcip_geo.countryName` are examined. If the detected login location is too far away (`distance_deviation` in miles) from that user's typical locations, an alert is triggered. The source host's reputation (`srcip_reputation`) is also checked. Map views of the Interflow include data points for the closest `typical` login locations for the user.

Location History Retention 5.4.0

To optimize location history tracking and improve detection accuracy, this alert type manages historical data. A location history retention mechanism provides the detection model with the last visit times and location, ensuring that more relevant data is retained for analysis and that noise from irrelevant historical locations is reduced. Users can receive alerts for locations they have visited in the past, if enough time has passed.

The last visited times are tracked for each location. If a user has not visited the location in a long time, another alert can be triggered, as follows:

- A user with only one location can get an alert on that location if three months have elapsed without another visit to that location.
- A user who visits lots of locations (defined as 10 unique locations in total and six locations in the last month (30 days)), can get an alert on that location after six months.
- If 50 users, sharing a tenant ID, visit a location, no further alerts for users with that tenant ID and location will be generated.

Remove Duplicate Login Events for Microsoft 365 and Entra ID 6.2.0

When both Microsoft 365 and Entra ID data sources report the same login ID, Stellar Cyber will prioritize Entra ID for alert generation of login location anomalies, provided there is no significant ingestion delay. If Stellar Cyber can correlate the two logins by their login ID, the Entra ID record will be sent. The Entra ID records have metadata related to MFA logins as well as specific Azure location enrichment, that the Microsoft 365 records do not have. The Entra ID record will still be prioritized even if it is not for MFA authentication.

However, if the Entra ID record is delayed, such as due to an ingestion delay, in order to maintain responsiveness, the Microsoft 365 record will be sent.

And if only Microsoft 365 is configured, those logs will continue to be used exclusively.

Using ASNs and User Agents to Improve Location Alert Triage 6.3.0

To improve triage of location-based alerts, Stellar Cyber will begin collecting user history for Autonomous System Number (ASN), ASN org, and user agents (if available from Microsoft 365), and then use this history to enrich alerts.

Enrichment fields are added to alerts detailing whether that user or tenant has seen that ASN, ASN org, or user agent before.

For the user and tenant scopes, the enriched fields are: `asn_seen_before`, `asn_org_seen_before`, and `user_agent_seen_before`.

The user scope has an additional enriched field: `user_start_time_ms`, which is the time that tracking ASN and user agents started, in UNIX milliseconds.

Using ASNs and user agents improves accuracy in anomaly detection and reduces false positives. You can also use the enriched fields for Automated Threat Hunting (ATH) or alert filtering.

User Process Usage Anomaly

A user who typically executes a small, consistent number of processes suddenly executed a new process. Investigate the process to see if it is benign. Check with the user to see if this process was expected.

XDR Kill Chain

- Kill Chain Stage: Persistent Foothold
- Tactic: XDR EBA (XTA0001)
- Technique: XDR Process Anomaly (XT1001)
- Tags: [Process Anomaly; User Behavior Analytics; Identity Detection]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `user_uncommon_process`.

Severity

10

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip_usersid
- user.name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip_usersid` — non-Windows source user ID

  or

- `user.identifier` — Windows source user ID

  
  The key field for this alert type can be either `srcip_usersid` or `user.identifier`, depending on the data feed.

- `hostip` — IP address of the host
- `hostip_host` — host name
- `srcip_username` — source user name
- `wineventlog_user.name` — source user name (Windows)
- `user.name` — source user name (Windows)
- `process_name` — name of the process
- `stability` — score measuring the time since the last new process was executed
- `days_stable` — time since the last new process was executed
- `diversity` — score measuring the number of processes that the user executed
- `child_count` — number of processes that the user executed

Use Case with Data Points

Looks for a user (`srcip_usersid` or `user.identifier` and a `srcip_username`) with a small number of processes (`diversity`, `child_count`) who also has not used a new process for a long time (`stability`, `days_stable`). If a new process (`process_name`) appears on a host (`srcip_host`) with this user and connects to another host (`dstip_host`), an alert is triggered.

The user is identified with the `scrip_userid` or `user.identifier` and `scrip_username` fields. The process is identified with the `process_name` field. The host on which the user is running the process is identified with the `srcip_host` field. The destination of the traffic generated by the process is identified with the `dstip_host` field. Stability is identified with the `stability` field, and diversity is identified with the `diversity` field.

Volume Shadow Copy Deletion via VssAdmin

The `vssadmin.exe` utility was used to delete the Shadow Copy on an endpoint. Ransomware and other malware do this to prevent system recovery. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Impact (TA0040)
- Technique: Inhibit System Recovery (T1490)
- Tags: [Malware; Ransomware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ransomware_volume_shadow_copy_deletion_via_vssadminedit`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- computer_name

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host where the Shadow Copy was deleted
- `hostip_host` — host name
- `process_name` — name of the executed process
- `event_data.CommandLine` — command line that is executed to delete the Shadow Copy

Use Case with Data Points

If `vssadmin.exe` is used to delete the Shadow Copy on an endpoint, an alert is triggered. The Interflow ibncludes the host IP address (`hostip`), process name (`process_name`), and command line (`event_data.CommandLine`).

Volume Shadow Copy Deletion via WMIC

The `wmic.exe` utility was used to delete the Shadow Copy on an endpoint. Ransomware and other malware do this to prevent system recovery. Check with the user.

XDR Kill Chain

- Kill Chain Stage: Exfiltration & Impact
- Tactic: Impact (TA0040)
- Technique: Inhibit System Recovery (T1490)
- Tags: [Malware; Ransomware]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `ransomware_volume_shadow_copy_deletion_via_wmicredit`.

Severity

80

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- hostip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `hostip` — IP address of the host where the Shadow Copy was deleted
- `hostip_host` — host name
- `process_name` — name of the executed process
- `event_data.CommandLine` — command line that is executed to delete the Shadow Copy

Use Case with Data Points

If `wmic.exe` is used to delete the Shadow Copy on an endpoint, an alert is triggered. The Interflow includes the host IP address (`hostip`), process name (`process_name`), and command line (`event_data.CommandLine`).

WAF Internal Attacker Anomaly

Internal web requests from a private IP address have been blocked/alerted by the Web Application Firewall (WAF). Investigate the source requester and ensure they are not compromised.

XDR Kill Chain

- Kill Chain Stage: Exploration
- Tactic: [Internal] XDR NBA (XTA0002)
- Technique: XDR WAF Anomaly (XT2009)
- Tags: [Internal; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `waf_internal_attacker`.

Severity

60

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- srcip

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `action` — status of web requests
- `event.severity_str` — severity level of web requests
- `event.uri` — URI of the web request
- `event.reason` — attack type (signature name)

The above fields are standardized to support a variety of WAFs. The original fields, listed below, remain in the F5 WAF Interflow record for backward compatibility.

List of F5 legacy fields

F5 Field Purpose	Original Stellar Cyber WAF Field Name	Standardized Stellar Cyber WAF Field Name
Signature ID	f5.sig_ids	event.sig_id
Signature name	f5.sig_names	event.reason
Staged signature ID	f5.staged_sig_ids	event.staged_sig_id
Staged signature ID	f5.staged_sig_names	event.staged_sig_name
Violation	f5.violations	event.violations
Sub violation	f5.sub_violations	event.sub_violations
Threat campaign	f5.threat_campaign_names	event.threat_campaign_names
Request status	f5.request_status	action
Severity	f5.severity	event.severity_str
Attack type	f5.attack_type	threat
Client IP	srcip	srcip service.origin.ip
Client Port	srcport	srcport service.origin.port
Service IP	dstip	dstip service.target.ip
Service Port	dstport	dstport service.target.port
Violation Details	f5.violation_details	event.description
Telemetry Event Category	f5.telemetryEventCategory	event.telemetry_event_category
url	f5.uri	event.uri
Web application name	f5.web_application_name	event.web_application_name

Use Case with Data Points

If web requests (f5.uri) from an internal IP address (srcip) to a web application (f5.web_application_name) have been blocked/alerted (f5.request_status) by the WAF, an alert is triggered. The Interflow includes the level of severity (f5.severity), the attack type (f5.attack_type), and the violation information (f5.violations), as well as signature name (f5.sig_names), staged signature name (f5.staged_sig_names), sub violation information (f5.sub_violations), and threat campaign name (f5.violation_details_xml.request-violations.violation.threat_campaign_data.threat_campaign_name), if applicable.

If web requests (`event.uri`) from an internal IP address (`srcip`) to a web application (`event.web_application_name`) have been blocked/alerted (`action`) by the WAF, an alert is triggered. The Interflow includes the level of severity (`event.severity_str`), the attack type (`threat`), and the violation information (`event.description`), as well as signature name (`event.reason`). If applicable for the WAF type, the Interflow also includes staged signature name (`event.staged_sig_id`), sub violation information (`event.sub_violations`), and threat campaign name (`event.threat_campaign.names`).

Ingestion Types Supported for this Alert

- F5 Big-IP Firewall
- F5 Silverline WAF
- Barracuda WAF
- AWS CloudWatch WAF
- Imperva Incapsula

WAF Rule Violation Anomaly

Web requests have been blocked/alerted by the Web Application Firewall (WAF) due to a surge in violations or violating a rule that is rarely invoked. Investigate the blocked/alerted web requests and ensure they are benign.

Refer to **Log Parser Ports**for the most current list of WAF parsers.

XDR Kill Chain

- Kill Chain Stage: Initial Attempts
- Tactic: [External] XDR NBA (XTA0002)
- Technique: XDR Rule Violation (XT2004)
- Tags: [External; Network Traffic Analysis]

Event Name

The `xdr_event.name` for this alert type in the Interflow data is `waf_rule_violation`.

Severity

50

Alert Suppression Fields

Alert suppression fields are attribute values that trigger deduplication of alerts sharing them.

- event.sig_id

Key Fields and Relevant Data Points

Key fields are meaningful and human-readable data points representing observable(s) being detected.

- `event.sig_id` — signature ID
- `srcip` — source IP address
- `srcip_host` — host name of corresponding source IP address
- `event.severity_str` — severity level of web requests
- `event.web_application_name` — web application name
- `event.uri` — URI of the web request
- `event.reason` — signature name
- `actual` — actual number of specific WAF violations in the period
- `typical` — typical number of specific WAF violations in the period

The above fields are standardized to support a variety of WAFs. The original fields, listed below, remain in the F5 WAF Interflow record for backward compatibility.

List of F5 legacy fields

F5 Field Purpose	Original Stellar Cyber WAF Field Name	Standardized Stellar Cyber WAF Field Name
Signature ID	f5.sig_ids	event.sig_id
Signature name	f5.sig_names	event.reason
Staged signature ID	f5.staged_sig_ids	event.staged_sig_id
Staged signature ID	f5.staged_sig_names	event.staged_sig_name
Violation	f5.violations	event.violations
Sub violation	f5.sub_violations	event.sub_violations
Threat campaign	f5.threat_campaign_names	event.threat_campaign_names
Request status	f5.request_status	action
Severity	f5.severity	event.severity_str
Attack type	f5.attack_type	threat
Client IP	srcip	srcip service.origin.ip
Client Port	srcport	srcport service.origin.port
Service IP	dstip	dstip service.target.ip
Service Port	dstport	dstport service.target.port
Violation Details	f5.violation_details	event.description
Telemetry Event Category	f5.telemetryEventCategory	event.telemetry_event_category

F5 Field Purpose	Original Stellar Cyber WAF Field Name	Standardized Stellar Cyber WAF Field Name
url	f5.uri	event.uri
Web application name	f5.web_application_name	event.web_application_name

Use Case with Data Points

If web requests (`event.uri`) to a web application (`event.web_application_name`) have been blocked/alerted (`action`) by the WAF due to violating certain rules, which include the level of severity (`event.severity_str`), the attack type (`threat`), and the violation information (`event.violations`). If the violations (`actual`) surge compared to the normal number of violations in a period (`typical`), an alert is triggered.

Ingestion Types Supported for this Alert

- F5 Big-IP Firewall
- F5 Silverline WAF
- Barracuda WAF
- AWS CloudWatch WAF
- Imperva Incapsula

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